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REFRIGERATOR AND METHOD FOR CONTROLLING REFRIGERATOR

Abstract

A refrigerator may include: a main body defining a storage chamber; a door configured to open and close the storage chamber; a refrigeration cycle device including a compressor and an evaporator and operable to cool the storage chamber; a thermoelectric element operable to cool the storage chamber; at least one sensor configured to produce sensor data associated with the refrigerator; and at least one processor configured to: operate the compressor to perform a refrigeration cycle based on a cooling condition being satisfied, start a cooling mode based on a defined condition associated with a time that the door is open being satisfied, and based on the cooling mode being started, obtain a predicted temperature value of the storage chamber from a temperature prediction model based on the sensor data produced by the at least one sensor, and operate the thermoelectric element to cool the storage chamber based on a difference between the predicted temperature value obtained from the temperature prediction model and a target temperature value being greater than a defined value.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This is a continuation application, under 35 U.S.C. § 111 (a), of International Application PCT/KR2025/099397, filed Feb. 14, 2025, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0032702, filed Mar. 7, 2024 and Korean Patent Application No. 10-2024-0072568, filed Jun. 3, 2024, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein in their entireties by reference.

TECHNICAL FIELD
[0002] The disclosure relates to a refrigerator equipped with a thermoelectric element and a refrigeration cycle device for cooling a storage and a method for controlling the refrigerator.
BACKGROUND ART
[0003] The refrigerator is a home appliance having a main body with storages and a cold air supply provided for supplying cold air into the storages to keep things fresh.
[0004] For the cold air supply of the refrigerator, a thermoelectric cooling device that causes heating and cooling actions through the Peltier effect may be used. The thermoelectric cooling device may include a thermoelectric element. The thermoelectric element may have a heater formed on one side and a cooler formed on the other side, and when a current is applied to the thermoelectric element, a heating action may occur in the heater and a heat absorption action may occur in the cooler.
[0005] The thermoelectric cooling device may be equipped with a heat sink, a cooling sink, a heat radiation fan, a cooling fan, a heat radiation duct and a cooling duct to increase cooling efficiency for the storage through the thermoelectric cooling device.
DISCLOSURE
Technical Problem
[0006] The disclosure provides a refrigerator with economic energy consumption and improved constant temperature performance and a method for controlling the refrigerator.
[0007] The disclosure provides a refrigerator and method for controlling the refrigerator, which may minimize temperature changes due to opening or closing of the door.
[0008] The disclosure provides a refrigerator and method for controlling the refrigerator, which may minimize temperature changes due to an object having large heat capacity.
[0009] The disclosure provides a refrigerator and method for controlling the refrigerator, which enhances constant temperature performance by using not only the current temperature of a storage but also a predicted temperature.
[0010] Technological objectives of the disclosure are not limited to what are mentioned above, and throughout the specification, it will be clearly appreciated by those of ordinary skill in the art that there may be other technological objectives unmentioned.

Technical Solution

[0011] In accordance with the present disclosure, a refrigerator may include: a main body defining a storage chamber; a door configured to open and close the storage chamber; a refrigeration cycle device including a compressor and an evaporator and operable to cool the storage chamber; a thermoelectric element operable to cool the storage chamber; at least one sensor configured to produce sensor data associated with the refrigerator; and at least one processor configured to: operate the compressor to perform a refrigeration cycle based on a cooling condition being satisfied, and start a cooling mode based on a defined condition associated with a time that the door is open being satisfied, based on the cooling mode being started, obtain a predicted temperature value of the storage chamber from a temperature prediction model based on the sensor data produced by the at least one sensor, and operate the thermoelectric element to cool the storage chamber based on a difference between the predicted temperature value obtained from the temperature prediction model and a target temperature value being greater than a defined value.

[0012] The at least one processor may be further configured to terminate the cooling mode by stopping operating the thermoelectric element in response to the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to a reference value after operating the thermoelectric element to cool the storage chamber.

[0013] The at least one processor may be further configured to terminate the cooling mode without operating the thermoelectric element based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to the defined value until the refrigeration cycle is performed a defined number of times.

[0014] The at least one processor may be further configured to determine the target temperature value based on a set temperature and the sensor data produced by the at least one sensor.

[0015] The at least one processor may be further configured to operate the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is being performed such that the thermoelectric element and the compressor are being operated together.

[0016] The at least one processor may be further configured to operate the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is not being performed such that only the thermoelectric element is being operated from among the thermoelectric element and the compressor.

[0017] The at least one processor may be further configured to: operate the thermoelectric element to cool the storage chamber based on a first control parameter in response to the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value, and while operating the thermoelectric element based on a second control parameter in response to a control condition having higher priority than the cooling mode being satisfied, even though the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value is greater than the defined value, keep operating the the thermoelectric element to cool the storage chamber based on the second control parameter.

[0018] The at least one processor may be further configured to: obtain the predicted temperature value of the storage chamber from the temperature prediction model at defined intervals, and change the defined intervals based on a temperature of the storage chamber.

[0019] The at least one processor is further configured to: obtain the predicted temperature value of the storage chamber from the temperature prediction model at defined intervals, and change the defined intervals based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.

[0020] The at least one processor may include: a first processor configured to control the refrigeration cycle device and the thermoelectric element; and a second processor configured to obtain the predicted temperature value of the storage chamber from the temperature prediction model; and the first processor may be further configured to send to the second processor an instruction to obtain the predicted temperature value from the temperature prediction model in response to the cooling mode being started, and the second processor may be further configured to: obtain the predicted temperature value from the temperature prediction model in response to the instruction of the first processor being received, and send the predicted temperature value obtained from the temperature prediction model to the first processor.

[0021] The at least one processor may be further configured to control a duty ratio of the thermoelectric element based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.

[0022] The at least one processor may be further configured to determine whether to operate the compressor based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.

[0023] The defined condition may include a cumulative time that the door is open exceeding a defined time, and the at least one processor may be further configured to initialize the cumulative time based on the cooling mode being started.

[0024] In accordance with the present disclosure, a method for controlling a refrigerator including a main body defining a storage chamber, a door configured to open and close the storage chamber, a refrigeration cycle device including a compressor and an evaporator and operable to cool the storage chamber, a thermoelectric element operable to cool the storage chamber, at least one sensor configured to produce sensor data associated with the refrigerator, and at least one processor, the method may include: by the at least one processor, operating the compressor to perform a refrigeration cycle based on a cooling condition being satisfied, and starting a cooling mode based on a defined condition associated with a time that the door is open being satisfied, and based on the cooling mode being started, obtaining a predicted temperature value of the storage chamber from a temperature prediction model based on the sensor data produced by the at least one sensor, and operating a thermoelectric element to cool the storage chamber based on a difference between the predicted temperature value obtained from the temperature prediction model and a target temperature value being greater than a defined value.

[0025] The method may further include: by the at least one processor, terminating the cooling mode by stopping operating the thermoelectric element in response to the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to a reference value after operating the thermoelectric element to cool the storage chamber.

[0026] The method may further include: by the at least one processor, terminating the cooling mode without operating the thermoelectric element based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to the defined value until the refrigeration cycle is performed a defined number of times.

[0027] The method may further include: by the at least one processor, determining the target temperature value based on a set temperature and the sensor data produced by the at least one sensor.

[0028] The operating the thermoelectric element comprises: operating the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is being performed such that the thermoelectric

element and the compressor are being operated together.
[0029] The operating the thermoelectric element comprises: operating the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is not being performed such that only the thermoelectric element is being operated from among the thermoelectric element and the compressor.
[0030] The method may further include: by the at least one processor, obtaining the predicted temperature value of the storage chamber from the temperature prediction model at defined intervals, and changing the defined intervals based on at least one of a temperature of the storage chamber or the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.

Description

DESCRIPTION OF DRAWINGS
[0031] FIG. **1** illustrates a refrigerator, according to an embodiment of the disclosure.
[0032] FIG. **2** illustrates a refrigerator with doors open, according to an embodiment of the disclosure.
[0033] FIG. **3** illustrates an upper portion of a storage of a refrigerator viewed from below, according to an embodiment of the disclosure.
[0034] FIG. **4** is a schematic side cross-sectional view of a refrigerator, according to an embodiment of the disclosure.
[0035] FIG. **5** is a cross-sectional view along line I-I of FIG. **2**.
[0036] FIG. **6** is an exploded view of a thermoelectric cooling device, according to an embodiment.
[0037] FIG. **7** is a block diagram illustrating an example of a configuration of a refrigerator, according to an embodiment.
[0038] FIG. **8** illustrates an example of a flowchart of a method by which a refrigerator cools a storage, according to an embodiment.
[0039] FIG. **9** illustrates an example of an operating condition of a thermoelectric element, according to an embodiment.
[0040] FIG. **10** is a flowchart illustrating an example of a method of controlling a refrigerator, according to an embodiment.
[0041] FIG. **11** illustrates an example in which a thermoelectric element is not operated even after a refrigerator starts a cooling mode, according to an embodiment.
[0042] FIG. **12** illustrates an example in which a compressor and a thermoelectric element are operated together when a refrigerator starts a cooling mode, according to an embodiment.
[0043] FIG. **13** illustrates an example in which only a thermoelectric element is operated among a compressor and the thermoelectric element when a refrigerator starts a cooling mode, according to an embodiment.
[0044] FIG. **14** illustrates an example in which a compressor is operated while a thermoelectric element is operating when a refrigerator starts a cooling mode, according to an embodiment.
MODES OF THE INVENTION
[0045] Embodiments and features as described and illustrated in the disclosure are merely examples, and there may be various modifications replacing the embodiments and drawings at the time of filing this application.
[0046] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to limit the disclosure.
[0047] For example, the singular forms “a”, “an” and “the” as herein used are intended to include

the plural forms as well, unless the context clearly indicates otherwise.
[0048] The terms "comprises" and/or "comprising," when used in this specification, represent the presence of stated features, integers, steps, operations, elements, components or combinations thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.
[0049] When an element is mentioned as being "connected to", "coupled to", "supported on" or "contacting" another element, it includes not only a case that the elements are directly connected to, coupled to, supported on or contact each other but also a case that the elements are connected to, coupled to, supported on or contact each other through a third element.
[0050] Throughout the specification, when an element is mentioned as being located "on" another element, it implies not only that the element is abut on the other element but also that a third element exists between the two elements.
[0051] The terms "forward or front", "rearward or back", "left", "right", "upper or up" or "lower or down" as herein used are defined with respect to the drawings, but the terms may not restrict the shapes and position of the respective components. For example, the front may be defined as +X direction and the back may be defined as −X direction. For example, with respect to the drawings, the right may be defined as +Y direction and the left may be defined as −Y direction. For example, with respect to the drawings, the upper direction may be defined as +Z direction and the lower direction may be defined as −Z direction.
[0052] The term including an ordinal number such as "first", "second", or the like is used to distinguish one component from another and does not restrict the former component.
[0053] Furthermore, the terms, such as "˜ part", "˜ block", "˜ member", "˜ module", etc., may refer to a unit of handling at least one function or operation. For example, the terms may refer to at least one process handled by hardware such as a field-programmable gate array (**9**)/application specific integrated circuit (ASIC), etc., software stored in a memory, or at least one processor.
[0054] An embodiment of the disclosure will now be described in detail with reference to accompanying drawings. Throughout the drawings, like reference numerals or symbols refer to like parts or components.
[0055] In an embodiment, a refrigerator may include a main body.
[0056] The main body may include insulation. The insulation may insulate inside and outside of the storage so that the temperature in the storage is maintained at a set suitable temperature without being influenced by external environments of the storage. In an embodiment, the insulation may include foam insulation such as polyurethane foam. In an embodiment, the insulation may include an additional vacuum insulation in addition to the foam insulation, or may include only the vacuum insulation instead of the foam insulation.
[0057] Various items such as foods, medicines, cosmetics, etc., may be stored in the storage, and the storage may be formed with one side open to put in or take out the items.
[0058] The refrigerator may include one or more storages. When there are two or more storages formed in the refrigerator, each storage may have a different use and may be maintained at a different temperature. For this, the storages may be separated by partition walls including insulation.
[0059] The storage chambers may be provided to be each maintained at a suitable temperature range, and may include a fridge, a freezer or a temperature-changing room classified by the use and/or the temperature range. The fridge may be maintained at a suitable temperature for keeping items refrigerated, and the freezer may be maintained at a suitable temperature for keeping items frozen. Refrigeration may refer to cooling the items to an extent that the items are not frozen, and for example, the fridge may be maintained at a range of 0 to 7 degrees Celsius above zero. Freezing may refer to freezing the items or cooling the items in a frozen state, and for example, the freezer may be maintained at a range of 20 to 1 degree Celsius below zero. The temperature-changing room may be used as one of the fridge or the freezer according to or regardless of the user's choice.

[0060] The storage chambers may be called many different names such as “veggie room”, “fresh room”, “cooling room” and “ice-making room” in addition to the names such as “fridge chamber”, “freezer chamber” and “temperature-changing room”, and the terms such as “fridge chamber”, “freezer chamber” and “temperature-changing room” need to be understood as encompassing storage chambers having respective uses and temperature ranges.
[0061] In an embodiment, the refrigerator may include at least one door arranged to open or close the one open side of the storage. A door may be equipped to open or close each of the one or more storages, or one door may be equipped to open or close multiple storages. The door may be rotationally or slidably installed at the front of the main body.
[0062] The door may be arranged to close the storage tight when closed. Like the main body, the door may include insulation to insulate the storage when closed.
[0063] In an embodiment, the door may include a door outer-plate that forms the front surface of the door, a door inner-plate that forms the rear surface of the door and faces the storage, an upper cap, a lower cap and door insulation provided inside of them.
[0064] A gasket may be arranged on edges of the door inner-plate to seal the storage by tightly contacting the front surface of the main body when the door is closed. The door inner-plate may include a dyke that protrudes rearward for a door basket that may keep items to be installed thereon.
[0065] In an embodiment, the door may include a door body and a front panel detachably coupled to the front side of the door body and forming the front of the door. The door body may include a door outer-plate that forms the front surface of the door body, a door inner-plate that forms the rear surface of the door body and faces the storage, an upper cap, a lower cap and door insulation provided inside of them.
[0066] The refrigerator may be distinguished according to the layout of the door(s) and storage(s) as a French door type, a side-by-side type, a bottom mounted freezer (BMF), a top mounted freezer (TMF) or a one-door refrigerator.
[0067] In an embodiment, the refrigerator may include a cold air supplier arranged to supply cold air into the storage.
[0068] The cold air supplier may include a machine, instrument, electronic device and/or system that combines them, which is able to produce and lead cold air to cool the storage.
[0069] In an embodiment, the cold air supplier may produce the cold air through a refrigeration cycle including processes of compression, condensation, expansion and evaporation of a refrigerant. For this, the cold air supplier may include a refrigeration cycle device having a compressor, a condenser, an expansion device and an evaporator that may operate the refrigeration cycle. In an embodiment, the cold air supplier may include semiconductors such as thermoelectric elements. The thermoelectric element may cool the storage chamber by heating and cooling actions through the Peltier effect.
[0070] In an embodiment, the refrigerator may include a machine room arranged for at least some parts belonging to the cold air supplier to be placed therein.
[0071] The machine room may be separated and insulated from the storage to prevent heat generated from the parts arranged in the machine room from being transferred to the storage. To emit heat from the parts arranged in the machine room, the inside of the machine room may be formed to connect to the outside of the main body.
[0072] In an embodiment, the refrigerator may include a dispenser arranged at the door to provide water and/or ice. The dispenser may be located at the door for the user to make access thereto without a need to open the door.
[0073] In an embodiment, the refrigerator may include an ice maker provided to produce ice. The ice maker may include an ice maker tray for storing water, an ice separator for separating ice from the ice maker tray, and an ice bucket for storing ice produced from the ice maker tray.
[0074] In an embodiment, the refrigerator may include a controller for controlling the refrigerator.

[0075] The controller may include a memory for storing or memorizing a program and/or data for controlling the refrigerator, and a processor for outputting control signals to control components such as the cold air supplier according to the program and/or data stored in the memory.
[0076] The memory stores or records various information, data, instructions, programs, etc., required for operation of the refrigerator. The memory may store temporary data that is generated while the control signals to control the components included in the refrigerator are being generated. The memory may include at least one or a combination of volatile memories or non-volatile memories.
[0077] The processor controls general operation of the refrigerator. The processor may control the components of the refrigerator by executing the program stored in the memory. The processor may include an extra neural processing unit (NPU) that performs operation of an artificial intelligence (AI) model. The processor may also include a central processing unit (CPU), a graphic processing unit (GPU), etc. The processor may generate control signals to control operation of the cold air supplier. For example, the processor may receive information about the temperature in the storage from a temperature sensor, and generate a refrigeration control signal to control an operation of the cold air supplier based on the temperature information of the storage.
[0078] Furthermore, the processor may process a user input to a user interface according to the program and/or data memorized/stored in the memory, and control operation of the user interface. The user interface may be provided by using an input interface and an output interface. The processor may receive the user input from the user interface. Furthermore, the processor may send a display control signal and image data for displaying an image on the user interface to the user interface in response to the user input.
[0079] The processor and the memory may be provided in one unit or separately. The processor may include one or more processors. For example, the processor may include a main processor and at least one sub-processor. The memory may include one or more memories.
[0080] In an embodiment, the refrigerator may include a processor and a memory for controlling all the components included in the refrigerator, or include a plurality of processors and a plurality of memories for controlling the components of the refrigerator, respectively. For example, the refrigerator may include a processor and a memory for controlling operation of the cold air supplier according to the output of the temperature sensor. The refrigerator may include another processor and another memory for controlling operation of the user interface according to the user input.
[0081] A communication module may communicate with an external device such as a server, a mobile device, another home appliance, etc., through a nearby access point (AP). The AP may connect a local area network (LAN) connected to the refrigerator or user device to a wide area network (WAN) connected to the server. The refrigerator or user device may be connected to the server through the WAN.
[0082] The input interface may include a key, a touch screen, a microphone, etc. The input interface may receive a user input and forward it to the processor.
[0083] The output interface may include a display, a speaker, etc. The output interface may output various notifications, alert, messages, information, etc., generated by the processor.
[0084] A working principle and embodiments of the disclosure will now be described with reference to accompanying drawings.
[0085] FIG. **1** illustrates a refrigerator, according to an embodiment of the disclosure. FIG. **2** illustrates a refrigerator with doors open, according to an embodiment of the disclosure. FIG. **3** illustrates an upper portion of a storage of a refrigerator viewed from below, according to an embodiment of the disclosure. FIG. **4** is a schematic side cross-sectional view of a refrigerator, according to an embodiment of the disclosure. FIG. **5** is a cross-sectional view along line I-I of FIG. **2**.
[0086] Referring to FIGS. **1** to **5**, a refrigerator **1** may include a main body **100**, storage chambers **11**, **12**, and **13** formed in the main body **100**, and doors **21**, **22**, **23**, and **24** arranged to open or close

the storages **11**, **12**, and **13**.

[0087] The main body **100** may include an inner case **170**, an outer case **180** coupled onto the outer side of the inner case **170**, and insulation **190** arranged between the inner case **170** and the outer case **180** (see FIG. **6**). The inner case **170** may define the storages **11**, **12** and **13**, and the outer case **180** may define an exterior of the main body **100**.

[0088] From another perspective, the main body **100** may include an upper wall **110**, a lower wall **120**, a left wall **130**, a right wall **140** and a rear wall **150**. The upper wall **110**, the lower wall **120**, the left wall **130**, the right wall **140** and the rear wall **150** may define the top surface, bottom surface, left surface, right surface and rear surface of the main body **100**, respectively.

[0089] The inner case **170**, the outer case **180** and the insulation **190** may define each of the upper wall **110**, the lower wall **120**, the left wall **130**, the right wall **140** and the rear wall **150**. For example, the top surface of the upper wall **110** may be defined by the outer case **180**, the bottom surface of the upper wall **110** may be defined by the inner case **170**, and the insulation **190** may be arranged within the upper wall **110**.

[0090] The storages **11**, **12** and **13** may accommodate items. The storages **11**, **12** and **13** may be formed to have open front to put in or take out the items. The main body **100** may include a horizontal partition wall **160** that divides a first storage **11** from a second storage **12** and a third storage **13**, and a vertical partition wall **161** that divides the second storage **12** from the third storage **13**. The first storage **11** may be arranged in an upper portion of the main body **100**, and the second storage **12** and the third storage **13** may be arranged in a lower portion of the main body **100**. The first storage **11** may be a fridge chamber; the second storage **12** may be a freezer chamber; the third storage **13** may be a temperature-changing chamber.

[0091] The doors **21**, **22**, **23** and **24** may open or close the storages **11**, **12** and **13**. The first door **21** and the second door **22** may open or close the first storage **11**, the third door **23** may open or close the second storage **12**, and the fourth door **24** may open or close the third storage **13**. The doors **21**, **22**, **22** and **24** may be rotationally coupled to the main body **100**.

[0092] The doors **21**, **22**, **22** and **24** may be rotationally coupled to the main body **100** by hinges. For example, the first door **21** and the second door **22** may be rotationally coupled to the main body **100** by hinges **31** arranged at the top of the main body **100** and hinges arranged in the middle of the main body **100**. The hinge **31** may include a hinge pin that vertically protrudes to form a rotation shaft of the door. The hinge **31** may be covered by a top cover **300** arranged to cover the top front portion of the main body **100**.

[0093] A rotation bar **40** may be arranged at one of the first door **21** and the second door **22** to cover a gap formed between the first door **21** and the second door **22** while the first door **21** and the second door **22** are closed. The rotation bar **40** may be rotationally arranged at one of the first door **21** and the second door **22**. The rotation bar **40** may have the shape of a bar formed to be long in the vertical direction. The rotation bar **40** may also be referred to as a pillar or a mullion.

[0094] A guide projection **46** may be arranged at the top of the rotation bar **40**, and a rotation guide **119** for guiding rotation of the guide projection **46** may be arranged at the top of the main body **100**.

[0095] The doors **21**, **22**, **23** and **24** may include gaskets **51**. The gaskets **51** may closely come into contact with the front surface of the main body **100** while the doors **21**, **22**, **23** and **24** are closed. The doors **21**, **22**, **23** and **24** may include dikes **52** that protrude rearward. Door racks **53** may be mounted on the dike **52** to store items. The rotation bar **40** may be rotationally installed at the dike **52**.

[0096] Although the number and layout of storages and the number and layout of doors were described above, there are no limitations on the number and layout of storages and the number and layout of doors of the refrigerator according to an embodiment of the disclosure.

[0097] The refrigerator **1** may include a thermoelectric cooling device **400** arranged to cool the storage **11**.

[0098] The thermoelectric cooling device **400** may be arranged above the storage **11** to cool the storage **11**. Specifically, the thermoelectric cooling device **400** may be arranged at the upper wall **110** of the main body **100**.
[0099] The thermoelectric cooling device **400** may include a thermoelectric element **530**. The thermoelectric element **530** may be a semiconductor device that uses the thermoelectric effect to convert thermal energy to electric energy and vice versa, and may also be referred to as a thermoelectric semiconductor device, a Peltier element, etc.
[0100] The thermoelectric element **530** includes a heater **531** and a cooler **532**. When a current is applied to the thermoelectric element **530**, a heat radiation action may occur in the heater **531** and a heat absorption action may occur in the cooler **532**. The thermoelectric element **530** may be shaped like a thin hexahedron. The heater **531** may be arranged on one side of the thermoelectric element **530** and the cooler **532** may be arranged on the other side.
[0101] The thermoelectric element **530** may be arranged at the upper wall **110** such that the heater **531** is directed upward of the thermoelectric element **530** and the cooler **532** is directed downward of the thermoelectric element **530**. In other words, the heater **531** may face outside of the main body **100** and the cooler **532** may face the inside of the storage **11**. Accordingly, the air heated by exchanging heat with the heater **531** may be discharged out of the main body **100** and the air cooled by exchanging heat with the cooler **532** may be supplied into the storage **11**.
[0102] The thermoelectric cooling device **400** may include a heat radiation sink **520** that comes into contact with the heater **531** to efficiently perform heat exchange between the heater **531** and the air outside the main body **100**.
[0103] The heat radiation sink **520** may be located outside the main body **100**. The heat radiation sink **520** may come into contact with the heater **531** to absorb heat from the heater **531** and emit heat to the outside of the main body **100**. The heat radiation sink **520** may also be referred to as a hot sink, a heat sink for heat dissipation, a hot heat sink, etc.
[0104] The heat radiation sink **520** may be formed of a metal with high heat conductivity. For example, the heat radiation sink **520** may be formed of aluminum or copper.
[0105] The heat radiation sink **520** may include a heat radiation sink base **521** that comes into contact with the heater **531**, and a plurality of heat radiation fins **525** protruding from the heat radiation sink base **521** to expand the heating surface. The plurality of heat sink fins **525** may protrude upward from the heat radiation sink base **521**.
[0106] The thermoelectric cooling device **400** may include a cooling sink **570** that comes into contact with the cooler **532** to efficiently perform heat exchange between the cooler **532** and the air inside the storage **11**.
[0107] The cooling sink **570** may be located in the storage **11**. The cooling sink **570** may cool the storage **11** by taking heat from the storage **11** and transferring the heat to the cooler **532**. The cooling sink **570** may also be referred to as a cold sink, a refrigeration sink, a cold heat sink, a cooling heat sink, etc.
[0108] The cooling sink **570** may be formed of a metal with high heat conductivity. For example, the cooling sink **570** may be formed of aluminum or copper.
[0109] The cooling sink **570** may include a cooling sink base **571** that comes into contact with the cooler **532**, and a plurality of cooling fins **575** protruding from the cooling sink base **571** to expand the heating surface. The plurality of cooling fins **525** may protrude downward from the cooling sink base **571**. The cooling sink base **571** and the plurality of cooling fins **575** may be integrally formed.
[0110] The thermoelectric cooling device **400** may include a heat radiation fan **600** that forces air to move around to efficiently perform heat exchange between the heat radiation sink **520** and the air outside the main body **100**.
[0111] The heat radiation fan **600** may be arranged to blow air to the heat radiation sink **520**. The heat radiation fan **600** may be located in a horizontal direction of the heat radiation sink **520**. The

heat radiation fan **600** may be arranged outside the main body **100**. The heat radiation fan **600** may be arranged on the top of the upper wall **110**.

[0112] The heat radiation fan **600** may be a centrifugal fan that draws air in the axial direction and discharges the air in the radial direction. The centrifugal fan may include a blower fan. A rotation shaft **610** of the heat radiation fan **600** may be arranged to be perpendicular to the top surface of the upper wall **110**.

[0113] The thermoelectric cooling device **400** may include a heat radiation duct **700** arranged to guide the air moving by the heat radiation fan **600**. The heat radiation duct **700** may draw in air from outside the main body **100** and guide the air to exchange heat with the heat radiation sink **520**, and discharge the air that has exchanged heat with the heat radiation sink **520** back to the outside of the main body **100**.

[0114] The heat radiation duct **700** may draw in air from an outer space above the main body **100**. The heat radiation duct **700** may discharge the air that has exchanged heat with the heat radiation sink **520** to the outer space above the main body **100**. The heat radiation fan **600** may be located in the heat radiation duct **700**. The heat radiation sink **520** may be located in the heat radiation duct **700**. The heat radiation duct **700** may be arranged on the top surface of the upper wall **110**.

[0115] The heat radiation duct **700** may include an outside air intake port **751** through which to draw in air from outside the main body **100** into the heat radiation duct **700**, and an outside air discharge port **782** through which to discharge the air that has exchanged heat with the heat radiation sink **520** to the outside of the main body **100**.

[0116] The thermoelectric cooling device **400** may include a cooling fan **800** that forces air to move to efficiently perform heat exchange between the cooling sink **570** and the air inside the storage **11**.

[0117] The cooling fan **800** may be arranged to blow air to the cooling sink **570**. The cooling fan **800** may be located in a horizontal direction of the cooling sink **570**. The cooling fan **800** may be arranged in the storage **11**. The cooling fan **800** may be arranged underneath the upper wall **110**.

[0118] The cooling fan **800** may be a centrifugal fan that draws air in the axial direction and discharges the air in the radial direction. A rotation shaft **810** of the cooling fan **800** may be arranged to be perpendicular to the bottom surface of the upper wall **110**.

[0119] The thermoelectric cooling device **400** may include a temperature sensor (hereinafter, a second defrost sensor) **112** for measuring temperature of the air cooled by the cooling fan **800**.

[0120] The second defrost sensor **112** may measure temperature of the cooling sink **570**. The measuring of the temperature of the cooling sink **570** may include measuring the temperature around the cooling sink **570** or measuring the temperature of the cooling sink **570** itself.

[0121] The second defrost sensor **112** may be arranged at the cooling sink **570**, or in a cooling duct **900**.

[0122] The thermoelectric cooling device **400** may include the cooling duct **900** arranged to guide the air moving by the cooling fan **800**. The cooling duct **700** may draw in air from inside the storage **11** and guide the air to exchange heat with the cooling sink **570**, and discharge the air that has exchanged heat with the cooling sink **570** back into the storage **11**.

[0123] The cooling fan **800** may be located in the cooling duct **900**. The cooling sink **570** may be located in the cooling duct **900**. The cooling duct **900** may be arranged underneath the upper wall **110**.

[0124] The cooling duct **900** may include an inside air intake port **991** through which to draw in air from inside the storage **11** into the cooling duct **900**, and an inside air discharge port **992** through which to discharge the air that has exchanged heat with the cooling sink **570** into the storage **11**.

[0125] Referring to FIG. **4**, the refrigerator **1** may include a refrigeration cycle device **450** (see FIG. **7**) to cool the storage through a refrigeration cycle. The refrigeration cycle device **450** may include a compressor **2**, a condenser (not shown), an expansion device (not shown) and an evaporator **3**. The evaporator **3** may be arranged behind the storages **12** and **13**.

[0126] The refrigerator **1** may include a temperature sensor (hereinafter, a first defrost sensor) **111**

for measuring temperature of the evaporator **3**.
[0127] The first defrost sensor **111** may measure the temperature of the evaporator **3**. The measuring of the temperature of the evaporator **3** may include measuring the temperature around the evaporator **3** or measuring the temperature of the evaporator **3** itself.
[0128] The first defrost sensor **111** may be arranged at the evaporator **3**, or in evaporator ducts **60** and **70**.
[0129] The refrigerator **1** may include the evaporator ducts **60** and **70** that guide cold air produced from the evaporator **3**. The first evaporator duct **60** may be arranged behind the second storage **12** and the third storage **13**. The second evaporator duct **70** may be arranged behind the first storage **11**.
[0130] The cold air produced from the evaporator **3** may be drawn into the first evaporator duct **60** by the evaporator fan **80**. The cold air drawn into the first evaporator duct **60** may be discharged into the second storage **12** or the third storage **13** through a cold air outlet (not shown) formed on the front. Furthermore, the cold air drawn into the first evaporator duct **60** may be guided into an internal flow path **78** of the second evaporator duct **70**. A damper **61** may be arranged in the first evaporator duct **60** to control the cold air in the first evaporator duct **60** to be supplied into the second evaporator duct **70**. A connection duct **90** may be arranged between the first evaporator duct **60** and the second evaporator duct **70** to connect the first evaporator duct **60** to the second evaporator duct **70**.
[0131] The cold air brought into the internal flow path **78** of the second evaporator duct **70** may be supplied into the first storage **11** through the cold air outlet **72** formed on the front of the second evaporator duct **70**.
[0132] However, unlike the aforementioned embodiment, the cold air produced from the evaporator **3** may be supplied directly into the second evaporator duct **70** without passing through the first evaporator duct **60**. Furthermore, the separate evaporator **3** may be arranged behind the first storage **11** to supply cold air into the second evaporator duct **70**.
[0133] As such, as the refrigerator **1** according to an embodiment of the disclosure may include the thermoelectric cooling device **400** and the refrigeration cycle device **450** for cooling the storage **11**, the method of supplying cold air into the storage **11** may include a first method of supplying cold air produced only by the thermoelectric cooling device **400**, a second method of supplying cold air produced only by the refrigeration cycle device **450**, and a third method of supplying cold air produced by both the thermoelectric cooling device **400** and the refrigeration cycle device **450**.
[0134] The refrigerator **1** may supply cold air into the storage **11** in a proper method according to external and internal conditions. For example, the refrigerator **1** may cool the storage **1** in one of the methods according to the temperature of a room where the refrigerator **1** is installed. Specifically, when the room temperature is higher than a defined temperature, so cooling by the refrigeration cycle has higher efficiency than by the thermoelectric cooling device **400**, the storage **11** may be cooled by the cold air produced only by the refrigeration cycle device **450**. On the other hand, when the room temperature is lower than the defined temperature, so cooling by the thermoelectric cooling device **400** has higher efficiency than by the refrigeration cycle device **450**, the storage **11** may be cooled by the cold air produced only by the thermoelectric cooling device **400**. The refrigerator **1** may operate only the thermoelectric cooling device **400** when there is a need to reduce noise. The refrigerator **1** may supply cold air produced by the thermoelectric cooling device **400** and cold air produced by the refrigeration cycle device **450** into the storage **11** at the same time when the storage **11** needs to be cooled rapidly.
[0135] As described above, the refrigerator **1** according to an embodiment of the disclosure may include the thermoelectric cooling device **400** and the refrigeration cycle device **450**, but is not limited thereto and the refrigerator may include only the thermoelectric cooling device **400**.
[0136] Although the thermoelectric cooling device **400** was described as being arranged at the upper wall **110** of the main body **100**, the location of the thermoelectric cooling device **400** is not

limited thereto.

[0137] In various embodiments, the thermoelectric cooling device **400** may be arranged at at least one of the upper wall **110**, the lower wall **120**, the left wall **130**, the right wall **140** and the rear wall **150**.

[0138] FIG. **6** is an exploded view of a thermoelectric cooling device, according to an embodiment.

[0139] Referring to FIG. **6**, the thermoelectric cooling device **400** may include a thermoelectric module **500**.

[0140] The aforementioned thermoelectric element **530**, the heat radiation sink **520**, and the cooling sink **570** may be assembled together to constitute the thermoelectric module **500**. In other words, the thermoelectric module **500** may include the thermoelectric element **530**, the heat radiation sink **520**, the cooling sink **570** and a module plate **550**.

[0141] The module plate **550** may serve as a frame of the thermoelectric module **500**. The module plate **550** may be formed of a resin material with low heat conductivity. The module plate **550** may keep a distance between the heat radiation sink **520** and the cooling sink **570**, and support the heat radiation sink **520** and the cooling sink **570**. The module plate **550** may be integrally formed with a fan case **650** which will be described later. However, it is also possible that the module plate **550** is arranged separately from the fan case **650**.

[0142] The module plate **550** may include a heat radiation sink support **552** for supporting the heat radiation sink **520**.

[0143] The module plate **550** may include a module plate opening **551**. The thermoelectric element **530** may be arranged inside the module plate opening **551**. The vertical length of the module plate opening **551** may be larger than the vertical length of the thermoelectric element **530**, and the thermoelectric element **530** may be arranged on the top of the module plate opening **551**. The reason why the thermoelectric element **530** is arranged on the top of the inside of the module plate opening **551** is that an amount of heat radiation of the thermoelectric element **530** is usually larger than an amount of heat absorption, that the thermoelectric element **530** being located on the top of the module plate opening **551** is advantageous to heat radiation of the heater **531**.

[0144] As the thermoelectric element **530** is arranged on the top of the module plate opening **551**, the cooling sink **570** may include a cooling conductor **574** that protrudes from the cooling sink base **571** to come into contact with the cooler **532** of the thermoelectric element **530**.

[0145] The thermoelectric module **500** may include an element insulator **540** that insulates the module plate **550** from the thermoelectric element **530**. The element insulator **540** may be arranged on the module plate opening **551** to prevent a side of the thermoelectric element **530** from contacting the module plate **550**. The element insulator **540** may include an element insulator opening **541** so that the thermoelectric element **530** may be accommodated in the element insulator opening **541**.

[0146] The thermoelectric module **500** may include a sink insulator **580** arranged between the module plate **550** and the cooling sink **570**. The sink insulator **580** may prevent heat transfer between the heat radiation sink **520** and the cooling sink **570** through the module plate **550**. The sink insulator **580** may include a sink insulator opening **581**. However, the sink insulator **580** may be omitted, and in this case, the heat radiation sink **520** may be supported on the top of the module plate **550** and the cooling sink **570** may be supported on the bottom of the module plate **550**.

[0147] The thermoelectric cooling device **400** may include the fan case **650** in which the heat radiation fan **600** is installed to guide air blown by the heat radiation fan **600**.

[0148] The fan case **650** may be formed integrally with or separately from the module plate **550**.

[0149] The fan case **650** may include a case bottom **660** on which the heat radiation fan **600** is rotationally installed, and a case scroll **670** extending upward from edges of the case bottom **660** to guide air blown from the heat radiation fan **600** toward the heat radiation sink **520**. The heat radiation fan **600** may be a centrifugal fan, which may be installed on the case bottom **660** so that the rotation shaft **610** is perpendicular to the case bottom **660**. Furthermore, the heat radiation fan

600 may be arranged so that the heat radiation sink **520** is located in a radial direction. With this structure, the overall vertical length of the thermoelectric cooling device **400** may be compact.

[0150] The case scroll **670** may be formed to surround the heat radiation fan **600**. The case scroll **670** may have a scroll opening **673** open toward the heat radiation sink **520**. The case scroll **670** may include a downstream end **671** of a rotation direction R of the heat radiation fan **600** and an upstream end **672** of the rotation direction R.

[0151] The fan case **650** may include a case guide **680** arranged to guide air moving to the vicinity of the downstream end **671** of the case scroll **670** from the heat radiation fan **600**.

[0152] The heat radiation sink **520** may include a plurality of heat radiation fins **525**. The plurality of heat sink fins **525** may protrude from the top **522** of the heat radiation sink base **521**. The plurality of heat sink fins **525** may protrude in a direction perpendicular to the top **522** of the heat radiation sink base **521**.

[0153] Heat radiation channels may be formed between the plurality of heat radiation fins **525**.

[0154] The heat radiation fan **600** may blow air toward the heat radiation sink **520**, and the air moved by the heat radiation fan **600** may exchange heat with the plurality of heat radiation fins **525** while passing through the heat radiation channels.

[0155] The cooling sink **570** may include a plurality of cooling fins **575**. The plurality of cooling fins **575** may be formed to extend in a direction parallel to the lower surface of the cooling sink base **571**.

[0156] Cooling channels may be formed between the plurality of cooling fins **575**.

[0157] The air moved by the cooling fan **800** may exchange heat with the plurality of cooling fins **575** while passing through the cooling channels.

[0158] FIG. **7** is a block diagram illustrating an example of a configuration of a refrigerator, according to an embodiment.

[0159] Referring to FIG. **7**, the refrigerator **1** according to an embodiment may include a sensor module **340**, the refrigeration cycle device **450**, the thermoelectric cooling device **400** and/or a controller **350**.

[0160] The sensor module **340** may include at least one sensor for collecting sensor data associated with the refrigerator **1**.

[0161] The sensor data associated with the refrigerator **1** may include data (e.g., temperature data, humidity data, image data, etc.) associated with internal environments (e.g., inside temperature, inside humidity, internal image, etc.) of the refrigerator **1** and/or data associated with external environments (e.g., outside temperature, outside humidity, proximity of the user) of the refrigerator **1**.

[0162] The sensor data associated with the refrigerator **1** may include data associated with states of components (e.g., the refrigeration cycle device **450**, the thermoelectric cooling device **400**, and the doors **21**, **22**, **23** and **24**) of the refrigerator **1**.

[0163] For example, the sensor data associated with the refrigerator **1** may include whether the refrigeration cycle device **450** operates, whether the thermoelectric cooling device **400** operates, or whether the doors **21**, **22**, **23** and **24** is open or closed.

[0164] In an embodiment, the sensor module **340** may include the first defrost sensor **111**, the second defrost sensor **112**, an indoor sensor **341**, an outdoor sensor **342** and/or a door sensor **343**.

[0165] An example of the sensor module **340** is not, however, limited thereto, and any sensor that is able to collect the sensor data associated with the refrigerator **1** may be employed in the sensor module **340**.

[0166] The first defrost sensor **111** may measure temperature of the evaporator **3**. The first defrost sensor **111** may send information about the temperature of the evaporator **3** to the controller **350**.

[0167] The second defrost sensor **112** may measure temperature of the cooling sink **570**. The second defrost sensor **112** may send information about the temperature of the cooling sink **570** to the controller **350**.

[0168] The indoor sensor **341** may measure temperature and/or humidity of the storage **11**, **12** or **13**. The indoor sensor **341** may send information about the temperature and/or humidity of the storage **11**, **12** or **13** to the controller **350**.
[0169] The outdoor sensor **342** may measure temperature and/or humidity outside the main body **100**. The outdoor sensor **342** may send information about the temperature and/or humidity outside the main body **100** to the controller **350**.
[0170] The door sensor **343** may detect whether the doors **21**, **22**, **23** and **24** are open or closed. The door sensor **343** may send information relating to opening or closing of the doors **21**, **22**, **23** and **24** to the controller **350**. For example, the door sensor **343** may send information about opening or closing of the first door **21** and/or the second door **22** that opens or closes the first storage **11** to the controller **350**.
[0171] The processor **351** of the controller **350** may control various components of the refrigerator **1** (e.g., the thermoelectric cooling device **400** and the refrigeration cycle device **450**) based on the information sent from the sensor module **340**.
[0172] The memory **352** of the controller **350** may store the information sent from the sensor module **340**.
[0173] The refrigeration cycle device **450** may cool the storages **11**, **12** and **13**.
[0174] The refrigeration cycle device **450** may include the compressor **2**, the condenser (not shown), the expansion device (not shown), and the evaporator **3**, and include an evaporator fan **80** for blowing the cold air produced from the evaporator **3** to the storages **11**, **12** and **13**.
[0175] The cold air produced from the evaporator **3** may be discharged into the second storage **12** or the third storage **13**, or may be supplied into the first storage **11** through the first evaporator duct **60**, the second evaporator duct **70** and the damper **61**.
[0176] In various embodiments, the controller **350** may control opening or closing the damper **61** during the operation of the refrigeration cycle device **450**, and thus open the damper **61** when cold air is to be supplied into the first storage **11** and close the damper **61** when there is no need to supply cold air into the first storage **11**.
[0177] The compressor **2** compresses the refrigerant, and supply the compressed refrigerant to the heat exchanger (e.g., the condenser (not shown), the expansion device (not shown) and the evaporator **3**).
[0178] The controller **350** may control the compressor **2** to adjust the temperature of the cold air produced from the evaporator **3**. For example, the controller **350** may control the compressor **2** to maintain the temperature measured by the indoor sensor **341** at a defined target temperature.
[0179] The controlling of the compressor **2** may include operating the compressor **2**. The operating of the compressor **2** may include controlling on/off of the compressor **2** or controlling an operating frequency of the compressor **2**.
[0180] The operating of the compressor **2** may include operating the evaporator fan **80** as well. The controller **350** may operate the compressor **2** along with the evaporator fan **80**.
[0181] The thermoelectric cooling device **400** may cool the storage **11**.
[0182] The thermoelectric cooling device **400** may include the thermoelectric element **530**, the heat radiation fan **600** and the cooling fan **800**.
[0183] When supplied with power, the thermoelectric element **530** may allow heat exchange between the cooling sink **570** and the heat radiation sink **520**. For example, the thermoelectric element **530** may convert electric energy to thermal energy so that heat radiation action occurs in the heater **531** and heat absorption action occurs in the cooler **532**.
[0184] When the heat radiation action occurs in the heater **531**, the air that has warmed by the heat radiation sink **520** that contacts the heater **531** may be discharged out of the main body **100** and the air that has cooled by the cooling sink **570** that contacts the cooler **532** may be supplied into the storage **11**.
[0185] The controller **350** may operate the thermoelectric element **530**. The operating of the

thermoelectric element **530** may include controlling a driving circuit for supplying power to the thermoelectric element **530**.

[0186] The operating of the thermoelectric element **530** may include turning on the thermoelectric element **530**. The operating of the thermoelectric element **530** may include turning on/off the thermoelectric element **530** with a defined duty ratio.

[0187] Turning on the thermoelectric element **530** may include supplying electric energy to the thermoelectric element **530**, i.e., suppling power to the thermoelectric element **530**. The supplying of the power to the thermoelectric element **530** may include applying a voltage and/or a current to the thermoelectric element **530**.

[0188] Turning off the thermoelectric element **530** may include not supplying electric energy to the thermoelectric element **530**, i.e., not suppling power to the thermoelectric element **530**. The not-supplying of the power to the thermoelectric element **530** may include not applying a voltage and/or a current to the thermoelectric element **530**.

[0189] When the thermoelectric element **530** is operated, the heat radiation sink **520** may come into contact with the heater **531** to absorb heat from the heater **531** and emit heat to the outside of the main body **100**.

[0190] When the thermoelectric element **530** is operated, the cooling sink **570** may cool the storage **11** by taking heat from the storage **11** and transferring the heat to the cooler **532**.

[0191] The heat radiation fan **600** may draw in air from outside the main body **100** and guide the air to exchange heat with the heat radiation sink **520**, and discharge the air that has exchanged heat with the heat radiation sink **520** back to the outside of the main body **100**.

[0192] The controller **350** may control the heat radiation fan **600**. The controlling of the heat radiation fan **600** may include controlling a fan motor of the heat radiation fan **600**. The controlling of the heat radiation fan **600** may include operating the heat radiation fan **600** and turning off the heat radiation fan **600**. The operating of the heat radiation fan **600** may include rotating the heat radiation fan **600** at a defined velocity. Turning off the heat radiation fan **600** may include stopping the rotation of the heat radiation fan **600**.

[0193] The fan motor of the heat radiation fan **600** may include a BLDC motor, speed of which is controllable.

[0194] As the air that has exchanged heat with the heat radiation sink **520** is moved with the operation of the heat radiation fan **600**, the heat radiation sink **520** may radiate heat fast. With the rapid heat radiation of the heat radiation sink **520**, heat radiation action in the heater **531** and heat absorption action in the cooler **532** may occur smoothly.

[0195] The cooling fan **800** may draw in air from inside the storage **11** and guide the air to exchange heat with the cooling sink **570**, and discharge the air that has exchanged heat with the cooling sink **570** back into the storage **11**.

[0196] The controller **350** may control the cooling fan **800**. The controlling of the cooling fan **800** may include controlling the fan motor of the cooling fan **800**. The controlling of the cooling fan **800** may include operating the cooling fan **800** and turning off the cooling fan **800**. The operating of the cooling fan **800** may include rotating the cooling fan **800** at a defined velocity. Turning off the cooling fan **800** may include stopping the rotation of the cooling fan **800**.

[0197] The fan motor of the cooling fan **800** may include a BLDC motor, speed of which is controllable.

[0198] As the air that has exchanged heat with the cooling sink **570** is moved around with the operation of the cooling fan **800**, the inside of the first storage **11** may be cooled fast. With the movement of the air that has exchanged heat with the cooling sink **570**, heat radiation action in the heater **531** and heat absorption action in the cooler **532** may occur smoothly.

[0199] The controller **350** may operate the cooling fan **800** and the heat radiation fan **600** when operating the thermoelectric element **530**.

[0200] In an embodiment, the controller **350** may operate the cooling fan **800** and the heat radiation fan **600** based on the thermoelectric element **530** being turned on. The operating of the cooling fan **800** and the heat radiation fan **600** based on the thermoelectric element **530** being turned on may include operating the cooling fan **800** and the heat radiation fan **600** after a lapse of a defined period of time after the thermoelectric element **530** is turned on, and/or operating the cooling fan **800** and the heat radiation fan **600** a defined time before the thermoelectric element **530** is turned on, and/or operating the cooling fan **800** and the heat radiation fan **600** when the thermoelectric element **530** is turned on.

[0201] In an embodiment, the controller **350** may turn off the cooling fan **800** and the heat radiation fan **600** based on the thermoelectric element **530** being turned off. The turning off of the cooling fan **800** and the heat radiation fan **600** based on the thermoelectric element **530** being turned off may include turning off the cooling fan **800** and the heat radiation fan **600** after a lapse of a defined period of time after the thermoelectric element **530** is turned off, and/or turning off the cooling fan **800** and the heat radiation fan **600** a defined time before the thermoelectric element **530** is turned off, and/or turning off the cooling fan **800** and the heat radiation fan **600** when the thermoelectric element **530** is turned off.

[0202] In the disclosure, the opposite concept of operating a specific component may be turning off the specific component or not operating the specific component.

[0203] In the disclosure, the operating of the refrigeration cycle device **450** may operating the compressor **2**.

[0204] In the disclosure, the operating of the thermoelectric cooling device **400** may include operating the thermoelectric element **530**.

[0205] The refrigerator **1** may include the communication interface **360** for communicating with an external device (e.g., a server or a user device) wiredly and/or wirelessly.

[0206] The communication interface **360** may include at least one processor **361** for controlling a communication module for transmitting or receiving data to or from an external device, and at least one memory **362** for storing a program and data for controlling the communication module.

[0207] The at least one memory **362** may store data required for various embodiments. The memory **362** may be implemented in the form of a memory embedded in or detachable from the refrigerator **1** depending on the data storage use. For example, data for operating the refrigerator **1** may be stored in the memory embedded in the refrigerator **1** and data for an extended function of the refrigerator **1** may be stored in the memory detachable from the refrigerator **1**. In the meantime, the memory embedded in the refrigerator **1** may be implemented with at least one of a volatile memory (e.g., a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous dynamic RAM (SDRAM), etc.) or a non-volatile memory (e.g., a one-time programmable read only memory (OTPROM), a programmable ROM (PROM), an erasable and programmable ROM (EPROM), an electrically erasable and programmable ROM (EEPROM), a mask ROM, a flash ROM, a flash memory (e.g., NAND flash or NOR flash), a hard drive or a solid state drive (SSD)). The memory detachable from the refrigerator **1** may be implemented in such a format as a memory card (e.g., compact flash (CF), secure digital (SD), micro-SD, mini-SD, extreme digital (xD), multi-media card (MMC), etc.) or an external memory (e.g., USB memory) connectable to a USB port.

[0208] In various embodiments, the at least one memory **352** or **362** may store a trained artificial intelligence (AI) model (e.g., a temperature prediction model), and the at least one processor **351** or **361** may use the trained AI model (e.g., the temperature prediction model) to obtain a predicted temperature value of the storage **11**.

[0209] In various embodiments, the trained AI model (e.g., the temperature prediction model) may be stored in an external device (e.g., a server).

[0210] The temperature prediction model may be trained to output a near-future temperature value (hereinafter, a predicted temperature value) of the storage **11** by using sensor data associated with the refrigerator **1** as input data. Herein, the term near future may refer to a time after a lapse of a

defined period of time (e.g., 10 minutes) from the current point in time.
[0211] The temperature prediction model may use the sensor data collected by the sensor module **340** to output the predicted temperature value of the storage **11**.
[0212] The predicted temperature value of the storage **11** may refer to a temperature value of the storage **11** predicted at a time after a lapse of the defined period of time (e.g., 10 minutes) from the current point in time.
[0213] In an embodiment, the refrigerator **1** may use the temperature prediction model to obtain the predicted temperature value of the storage **11**. In an embodiment, the refrigerator **1** may use the temperature prediction model only when a particular condition is satisfied, to obtain the predicted temperature value of the storage **11**.
[0214] The controller **350** and the communication interface **360** of the refrigerator **1** may be connected wiredly or wirelessly. For example, the processor **351** of the controller **350** and the processor **361** of the communication interface **360** may transmit or receive various information, commands or instructions to or from each other wiredly or wirelessly.
[0215] In an embodiment, when the temperature prediction model is stored in the memory **352** of the controller **350** and the processor **351** of the controller **350** may perform an operation of an AI model, the processor **351** may execute the temperature prediction model stored in the memory **352** and input sensor data collected from the sensor module **340** to the temperature prediction model to obtain a predicted temperature value of the storage **11**.
[0216] In the meantime, as the processor **351** of the controller **350** is a component for controlling general components of the refrigerator **1** (e.g., the thermoelectric cooling device **400** and the refrigeration cycle device **450**), data processing load may be too heavy for the processor **351** of the controller **350** to execute the temperature prediction model.
[0217] In an embodiment, the temperature prediction model may be stored in the memory **362** of the communication interface **360**, and the processor **361** of the communication interface **360** may perform an operation of the AI model. In this case, the processor **351** of the controller **350** may instruct the processor **361** of the communication interface **360** to perform the temperature prediction model, and in response to receiving the instruction, the processor **361** of the communication interface **360** may input the sensor data collected from the sensor module **340** to the temperature prediction model to obtain a predicted temperature value of the storage **11** and send this to the processor **351** of the controller **350**.
[0218] To instruct to perform the temperature prediction model, the processor **351** of the controller **350** may send the sensor data collected by the sensor module **340** to the processor **361** of the communication interface **360**.
[0219] In the disclosure, by using the processor **361** of the communication interface **360** to execute the temperature prediction model, data processing load of the processor **351** of the controller **350** may be reduced.
[0220] In the meantime, according to various embodiments, the temperature prediction model may be stored in an external device and the external device may perform an operation of the AI model. In this case, the processor **351** of the controller **350** may control the communication interface **360** to instruct the external device to perform the temperature prediction model, and in response to receiving the instruction, the external device may input the sensor data collected from the sensor module **340** to the temperature prediction model to obtain a predicted temperature value of the storage **11** and send this to the processor **351** of the controller **350**.
[0221] When instructing to perform the temperature prediction model, the processor **351** of the controller **350** may send the sensor data collected by the sensor module **340** to the external device through the communication interface **360**.
[0222] In an embodiment, to reduce the data processing load, the refrigerator **1** may execute the temperature prediction model at defined intervals when a defined condition is satisfied (e.g., when a cooling mode is started).

[0223] The communication interface **360** may transmit or receive data to or from an external device (e.g., a server or a user device). For this, the communication interface **360** may support establishment of a direct (e.g., wired) communication channel or a wireless communication channel between external devices, and communication through the established communication channel. In an embodiment, the communication interface **360** may include a wireless communication module (e.g., a cellular communication module, a short-range wireless communication module or a GNSS communication module) or a wired communication module (e.g., a LAN communication module or a power-line communication module). A corresponding one of the communication modules may communicate with an external device over a first network (e.g., a short-range communication network such as bluetooth, Wi-Fi direct or IrDA) or a second network (e.g., a remote communication network such as a legacy cellular network, a 5G network, a next generation communication network, the Internet, or a computer network (e.g., a LAN or WAN). These various types of communication modules may be integrated into a single component (e.g., a single chip) or implemented as a plurality of separate components (e.g., a plurality of chips).
[0224] The short-range communication module may include a bluetooth communication module, a BLE communication module, an NFC module, a WLAN, e.g., Wi-Fi, communication module, a Zigbee communication module, an IrDA communication module, a WFD communication module, an UWB communication module, an Ant+ communication module, a uWave communication module, etc., without being limited thereto.
[0225] The long-range communication module may include a communication module for performing various types of long-range communication and include a mobile communication interface. The mobile communication interface transmits and receives RF signals to and from at least one of a base station, an external terminal, or a server in a mobile communication network.
[0226] In an embodiment, the communication interface **360** may communicate with the external device through a nearby access point (AP). The AP may connect a local area network (LAN) connected to the refrigerator **1** to a wide area network (WAN) connected to the server. The refrigerator **1** may be connected to the server through the WAN.
[0227] The refrigerator **1** may receive various signals (e.g., remote instructions) from the external device through the communication interface **360**.
[0228] The refrigerator **1** may transmit various signals to the external device through the communication interface **360**.
[0229] The controller **350** may include the at least one processor **351** for controlling operation of the refrigerator **1**, and at least one memory **352** for storing a program and data for controlling the operation of the refrigerator **1**.
[0230] The at least one memory **352** may store data required for various embodiments. The memory **352** may be implemented in the form of a memory embedded in or detachable from the refrigerator **1** depending on the data storage use. For example, data for operating the refrigerator **1** may be stored in the memory embedded in the refrigerator **1** and data for an extended function of the refrigerator **1** may be stored in the memory detachable from the refrigerator **1**. In the meantime, the memory embedded in the refrigerator **1** may be implemented with at least one of a volatile memory (e.g., a dynamic random access memory (DRAM), a static RAM (SRAM), or a synchronous dynamic RAM (SDRAM), etc.) or a non-volatile memory (e.g., a one-time programmable read only memory (OTPROM), a programmable ROM (PROM), an erasable and programmable ROM (EPROM), an electrically erasable and programmable ROM (EEPROM), a mask ROM, a flash ROM, a flash memory (e.g., NAND flash or NOR flash), a hard drive or a solid state drive (SSD)). The memory detachable from the refrigerator **1** may be implemented in such a format as a memory card (e.g., compact flash (CF), secure digital (SD), micro-SD, mini-SD, extreme digital (xD), multi-media card (MMC), etc.) or an external memory (e.g., USB memory) connectable to a USB port. The at least one processor **351** controls general operation of the refrigerator **1**. Specifically, the at least one processor **351** may be connected to the respective

components of the refrigerator **1** (e.g., the sensor module **340**, the refrigeration cycle device **450**, the thermoelectric cooling device **400** and the communication interface **360**) to control general operation of the refrigerator **1**. For example, the at least one processor **351** may be electrically connected to the memory **352** to control general operation of the refrigerator **1**. The processor **351** may include one or more processors.

[0231] The at least one processor **351** may execute at least one instruction stored in the memory **352** to perform operation of the refrigerator **1** according to various embodiments.

[0232] The at least one processor **351** may include one or more of a central processing unit (CPU), a graphics processing unit (GPU), an accelerated processing unit (APU), a many integrated core (MIC), a digital signal processor (DSP), a neural processing unit (NPU), a hardware accelerator or a machine learning accelerator. The at least one processor **351** may control one or any combination of the other components of the refrigerator **1**, and perform a communication related operation or data processing. The at least one processor **351** may execute at least one program or instruction stored in the memory **352**. For example, the at least one processor **351** may execute the at least one instruction stored in the memory **352** to perform a method according to at least one embodiment of the disclosure.

[0233] In an embodiment, the controller **350** may perform a cooling operation in various methods.

[0234] In an embodiment, the controller **350** may perform a cooling operation by operating only the thermoelectric element **530** among the compressor **2** and the thermoelectric element **530** to supply cold air produced only by the thermoelectric cooling device **400** to the storage **11**.

[0235] In an embodiment, the controller **350** may perform a cooling operation by operating only the compressor **2** among the compressor **2** and the thermoelectric element **530** to supply cold air produced only by the refrigeration cycle device to the storage **11**.

[0236] In an embodiment, the controller **350** may perform a cooling operation by operating both the compressor **2** and the thermoelectric element **530** to supply both cold air produced by the thermoelectric cooling device **400** and cold air produced by the refrigeration cycle device to the storage **11**.

[0237] In an embodiment, the controller **350** may perform a refrigeration cycle by operating the compressor **2** based on current temperature of the storage **11**, **12** or **13** and a target temperature of the storage **11**, **12** or **13**.

[0238] For example, the controller **350** may perform a refrigeration cycle by operating the compressor **2** based on the temperature of the first storage **11** being higher than the target temperature of the first storage **11** (hereinafter, a first target temperature). In this case, the controller **350** may open the damper **61**.

[0239] In an embodiment, the controller **350** may determine the first target temperature based on a set temperature of the first storage **11** (hereinafter, a first set temperature). The set temperature may refer to a temperature that may be set by the user.

[0240] For example, the controller **350** may determine the first target temperature by correcting the first set temperature based on the sensor data (e.g., temperature data outside the refrigerator, humidity data outside the refrigerator, etc.) collected from the sensor module **340**. Accordingly, the first target temperature may be similar or equal to the first set temperature.

[0241] The controller **350** may terminate the refrigeration cycle by stopping operation of the compressor **2** based on the temperature of the first storage **11** falling to or below the first target temperature.

[0242] In another example, the controller **350** may perform a refrigeration cycle by operating the compressor **2** based on the temperature of the second storage **12** being higher than a target temperature of the second storage **12** (hereinafter, a second target temperature). In this case, the controller **350** may open or close the damper **61** according to the temperature of the first storage **11**.

[0243] In an embodiment, the controller **350** may determine the second target temperature based on a set temperature of the second storage **12** (hereinafter, a second set temperature).

[0244] For example, the controller **350** may determine the second target temperature by correcting the second set temperature based on the sensor data collected from the sensor module **340**. Accordingly, the second target temperature may be similar or equal to the second set temperature.
[0245] The controller **350** may terminate the refrigeration cycle by stopping operation of the compressor **2** based on the temperature of the second storage **11** falling to or below the second target temperature.
[0246] In the disclosure, the refrigeration cycle may refer to a period between when an operation of the compressor **2** is started and when the operation of the compressor **2** is terminated.
[0247] In the disclosure, the number of times that the refrigerator **1** performs the refrigeration cycle may refer to the number of times that the operation of the compressor **2** is stopped after the operation of the compressor **2** is started.
[0248] In an embodiment, an operating condition of the compressor **2** and an operating condition of the thermoelectric element **530** may be independent from each other.
[0249] According to the traditional technology, the refrigerator operates the compressor only when the temperature of the storage rises to the target temperature or higher to perform the refrigeration cycle. The performing of the refrigeration cycle only when the temperature of the storage rises to the target temperature or higher may cause the temperature of the storage to be far away from the target temperature when a rapid temperature rise in the storage is expected due to having an object with high heat capacity in the storage or due to a long door opening time.
[0250] To solve this problem, the refrigerator **1** according to an embodiment of the disclosure may operate the thermoelectric element **530** and optionally, further operate the compressor **2** when a rapid temperature rise in the storage **11** is expected by using the temperature prediction model, thereby preventing the rapid temperature rise in the storage **11**.
[0251] In the meantime, the refrigerator **1** according to an embodiment may include various components other than the aforementioned components. For example, the refrigerator **1** may include a user interface device (e.g., a display, an input device, a speaker, etc.) for receiving a user input and providing various information for the user to interact with the user.
[0252] FIG. **8** illustrates an example of a flowchart of a method by which a refrigerator cools a storage, according to an embodiment.
[0253] Referring to FIG. **8**, the controller **350** may determine whether the temperature of the storage **11**, **12** or **13** satisfies a first cooling condition, in **1100**. The temperature of the storage **11**, **12** or **13** may refer to temperature in the storage **11**, **12** or **13** at the current point in time measured by the indoor sensor **341**.
[0254] The first cooling condition may be referred to as a temperature condition in that it is a condition associated with the temperature in the storage **11**, **12** or **13**.
[0255] The first cooling condition may include temperature in the storage **11**, **12** or **13** exceeding a target temperature.
[0256] The controller **350** may determine that the first cooling condition is satisfied based on the temperature in the storage **11**, **12** or **13** exceeding the target temperature.
[0257] For example, the controller **350** may determine that the first cooling condition is satisfied based on the temperature in the first storage **11** exceeding first target temperature. The controller **350** may determine that the first cooling condition is satisfied based on the temperature in the second storage **12** exceeding second target temperature. The controller **350** may determine that the first cooling condition is satisfied based on the temperature in the third storage **13** exceeding third target temperature.
[0258] The first cooling condition may further include a rapid cooling condition.
[0259] The controller **350** may determine whether the temperature of the storage **11**, **12** or **13** satisfies the rapid cooling condition, in **1110**.
[0260] The rapid cooling condition may include the temperature of the storage **11**, **12** or **13** exceeding the target temperature, and a difference between the temperature of the storage **11**, **12** or

13 and the target temperature being larger than a defined value (e.g., 3° C.).
[0261] For example, the controller **350** may determine that the rapid cooling condition is satisfied based on the temperature of the first storage **11** exceeding a first target temperature and a difference between the temperature of the first storage **11** and the first target temperature being larger than a first defined value. The controller **350** may determine that the rapid cooling condition is satisfied based on the temperature of the second storage **12** exceeding a second target temperature and a difference between the temperature of the second storage **12** and the second target temperature being larger than a second defined value. The controller **350** may determine that the rapid cooling condition is satisfied based on the temperature of the third storage **13** exceeding a third target temperature and a difference between the temperature of the third storage **13** and the third target temperature being larger than a third defined value.
[0262] The controller **350** may perform only the refrigeration cycle in **1130** based on the first cooling condition of the storage **11**, **12** or **13** being satisfied in **1100** and the rapid cooling condition not being satisfied in **1100**.
[0263] The performing of only the refrigeration cycle may include operating only the compressor **2** among the compressor **2** and the thermoelectric element **530**.
[0264] Based on the rapid cooling condition of the storage **11**, **12** or **13** being satisfied in **1110**, the controller **350** may perform the refrigeration cycle and operate the thermoelectric element **530** in **1120**.
[0265] The performing of the refrigeration cycle and the operating of the thermoelectric element **530** may include operating both the compressor **2** and the thermoelectric element **530**.
[0266] The controller **350** may determine whether the temperature of the storage **11**, **12** or **13** satisfies a cooling termination condition, in **1140**.
[0267] The cooling termination condition may include the temperature in the storage **11**, **12** or **13** falling to the target temperature.
[0268] Based on the cooling termination condition of the storage **11**, **12** or **13** being satisfied in **1140**, the controller **350** may terminate cooling of the storage **11**, **12** or **13**.
[0269] The terminating of the cooling of the storage **11**, **12** or **13** may include terminating the refrigeration cycle by stopping the operation of the compressor **2**, and optionally, stopping the operation of the thermoelectric element **530**.
[0270] The controller **350** may stop operating the compressor **2** and the thermoelectric element **530** based on the cooling termination condition being satisfied during operation **1120**.
[0271] The controller **350** may stop operating the compressor **2** based on the cooling termination condition being satisfied during operation **1130**.
[0272] As such, the first cooling condition is associated with the current temperature of the storage **11**, **12** or **13**. Hence, controlling the compressor **2** and the thermoelectric element **530** based only on the first cooling condition may make it vulnerable to the rapid temperature change of the storage.
[0273] The controller **350** may determine whether a second cooling condition is satisfied, in **1200**.
[0274] The second cooling condition is different from the first cooling condition, and may be referred to as a constant temperature condition in that it is a condition to minimize the temperature change of the storage **11**, **12** or **13**.
[0275] The second cooling condition may include a first condition for operating the thermoelectric element **530**, and optionally include a second condition for operating the compressor **2**. The first and second conditions may be independent from each other.
[0276] The first condition may be referred to as an operation condition (or control condition) for the thermoelectric element **530**, and the second condition may be referred to as an operation condition (or control condition) for the compressor **2**.
[0277] FIG. **9** illustrates an example of an operating condition of a thermoelectric element, according to an embodiment.

[0278] Referring to FIG. **9**, the first condition may include a plurality of first conditions each having priority.
[0279] The plurality of first conditions may each include priority and a control parameter. The control parameter may include an output voltage and/or an on/off duty ratio of the thermoelectric element **530**.
[0280] The plurality of first conditions may each include a start condition and a termination condition. The controller **350** may control the thermoelectric element **530** based on the corresponding control parameter when the start condition is satisfied, and stop controlling the thermoelectric element **530** when the termination condition is satisfied.
[0281] The controlling of the thermoelectric element **530** may include not only operating the thermoelectric element **530** but also maintaining an off state of the thermoelectric element **530**.
[0282] In an embodiment, the first condition may include receiving a user command to operate the thermoelectric element **530**.
[0283] On receiving the user command to operate the thermoelectric element **530**, the controller **350** may control the thermoelectric element **530** with a control parameter according to a user setting. When receiving the user command to operate the thermoelectric element **530**, the controller **350** may control the thermoelectric element **530** with a control parameter according to the user setting until a user command to stop operating the thermoelectric element **530** is received or a defined period of time elapses.
[0284] In an embodiment, the first condition may include condition M**1**. The condition M**1** may be a condition associated with a cooling mode as will be described later.
[0285] For example, a start condition of the condition M**1** may include the difference between an expected temperature value of the storage **11** and a target temperature value being larger than a defined value after the start of the cooling mode.
[0286] The cooling mode will be described in detail later.
[0287] A termination condition of the condition M**1** may include the difference between an expected temperature value of the storage **11** and the target temperature value falling to or below a reference value after the thermoelectric element **530** is operated.
[0288] The controller **350** may control the thermoelectric element **530** based on a control parameter M**2** based on the condition M**1** being satisfied.
[0289] In an embodiment, the first condition may include condition L**1** having higher priority than the condition M**1**.
[0290] In an embodiment, while the thermoelectric element **530** is being controlled in response to a first condition having high priority being satisfied, even when a first condition with low priority is satisfied, the thermoelectric element **530** may not be controlled with a control parameter corresponding to the first condition having the low priority.
[0291] Specifically, while the thermoelectric element **530** is being controlled with control parameter L**2** in response to the condition L**1** being satisfied, even when the condition M**1** having low priority is satisfied, the controller **350** may maintain to control the thermoelectric element **530** with control parameter L**2** without controlling the thermoelectric element **530** with control parameter M**2**.
[0292] The condition L**1** may include initial installation conditions, and various conditions associated with the sensor data collected by the sensor module **340**.
[0293] For example, a start condition of the condition L**1** may include a lapse of a certain period of time after the refrigerator **1** is powered on. A termination condition of the condition L**1** may include the temperature of the storage **11** falling to or below a defined temperature. When the start condition of the condition L**1** includes a lapse of the certain period of time after the refrigerator **1** is powered on, output voltage (or duty ratio) L**2** may have a relatively larger value than 0.
[0294] While the thermoelectric element **530** is being controlled with control parameter L**2** based on the condition L**1** being satisfied, even when the condition Ml having a priority lower than

condition L**1** is satisfied, the controller **350** may control the thermoelectric element **530** based on the control parameter L**2**.

[0295] In another example, the condition L**1** may include various conditions in which efficiency is low even when the thermoelectric element **530** operates. In this case, the output voltage (or duty ratio) L**2** of the condition L**1** may be set to 0.

[0296] When the condition in which the efficiency is low even when the thermoelectric element **530** operates is satisfied, the controller **350** may not operate the thermoelectric element **530** even when the condition M**1** having lower priority than the condition L**1** is satisfied.

[0297] A start condition of the condition in which the efficiency is low even when the thermoelectric element **530** operates may include, for example, a condition in which a certain period of time does not elapse right after the refrigerator **1** is powered on, a condition in which a temperature lower than a defined temperature is detected by the second defrost sensor **112** for a certain period of time, a condition in which an error of the refrigeration cycle device **450** is detected, a condition in which opening of the first door **21** or the second door **22** is continuously detected for a certain period of time (e.g., 5 minutes), and/or a condition in which the temperature outside the refrigerator is equal to or higher than a defined temperature (e.g., 39° C.).

[0298] A termination condition of the condition in which the efficiency is low even when the thermoelectric element **530** operates may include, for example, a condition in which a certain period of time elapses right after the refrigerator **1** is powered on, a condition in which a temperature higher than the defined temperature is detected by the second defrost sensor **112**, a condition in which an error of the refrigeration cycle device **450** is not detected, a condition in which closing of the first door **21** or the second door **22** is detected, and/or a condition in which the temperature outside the refrigerator is lower than the defined temperature (e.g., 39° C.). The condition L**1** may include various conditions apart from the aforementioned conditions.

[0299] In an embodiment, the first condition may include condition N**1** having a lower priority than the condition M**1**.

[0300] In various embodiments, the condition N**1** may include a condition regarding temperature and humidity outside the refrigerator.

[0301] For example, the condition N**1** may include a condition in which the temperature outside the refrigerator belongs to a first range and humidity outside the refrigerator is equal to or higher than defined humidity, a condition in which the temperature outside the refrigerator belongs to a second range lower than the first range and humidity outside the refrigerator is equal to or higher than defined humidity, and/or a condition in which the temperature outside the refrigerator belongs to a third range lower than the first range and humidity outside the refrigerator is equal to or higher than defined humidity.

[0302] An output voltage or duty ratio of the control parameter N**2** corresponding to the condition in which the temperature outside the refrigerator belongs to the first range and the humidity outside the refrigerator is equal to or higher than the defined humidity may be higher than an output voltage or duty ratio of the control parameter N**2** corresponding to the condition in which the temperature outside the refrigerator belongs to the second range and the humidity outside the refrigerator is equal to or higher than the defined humidity.

[0303] According to the disclosure, the thermoelectric element **530** may operate according to the first condition no matter whether the compressor **2** operates.

[0304] The second condition may be an operating condition of the compressor **2** regardless of the first cooling condition, and may include a plurality of second conditions. The second condition may include a start condition and a termination condition. The second condition may include a control parameter. The control parameter may include an operating frequency of the compressor **2** and whether to open or close the damper **61**.

[0305] In an embodiment, the second condition may be associated with a predicted temperature value of the second storage **12**, and optionally associated with a predicted temperature value of the first storage **11**.

[0306] The controller **350** may operate the compressor **2** in response to the start condition of the second condition being satisfied, and stop operating the compressor **2** in response to the termination condition of the second condition being satisfied.

[0307] The controller **350** may perform a hybrid cooling operation based on the second cooling condition being satisfied, in **1210**.

[0308] The hybrid cooling operation may include operating the thermoelectric element **530** based on the start condition of the first condition being satisfied, operating the compressor **2** based on the start condition of the second condition being satisfied, and operating both the compressor **2** and the thermoelectric element **530** based on the start conditions of the first and second conditions being both satisfied.

[0309] The controller **350** may operate the thermoelectric element **530** based on the start condition of the first condition being satisfied. The controller **350** may operate the compressor **2** based on the start condition of the second condition being satisfied.

[0310] Based on a second cooling termination condition being satisfied in **1220**, the controller **350** may terminate the hybrid cooling operation in **1230**.

[0311] The second cooling termination condition may correspond to the termination condition of the first condition and/or the termination condition of the second condition.

[0312] The controller **350** may stop operating the thermoelectric element **530** based on the termination condition of the first condition being satisfied. The controller **350** may stop operating the compressor **2** based on the termination condition of the second condition being satisfied.

[0313] According to the disclosure, provided are the refrigerator **1** for performing the hybrid cooling operation according to whether not only a temperature condition but also a constant temperature condition are satisfied, and a method for controlling the refrigerator **1**.

[0314] A cooling mode associated with the condition M**1** will now be described in detail.

[0315] FIG. **10** is a flowchart illustrating an example of a method of controlling a refrigerator, according to an embodiment.

[0316] Referring to FIG. **10**, the controller **350** may determine whether a defined condition associated with opening of the door **21**, **22**, **23** or **24** is satisfied, in **2100**.

[0317] The defined condition associated with opening of the door **21**, **22**, **23** or **24** may include a defined condition associated with opening of the first door **21** and/or the second door **22** which opens or closes the first storage **11**.

[0318] The defined condition associated with opening of the door **21**, **22**, **23** or **24** may include defined events in which a rapid temperature change of the first storage **1** is expected.

[0319] For example, the defined condition associated with opening of the door **21** or **22** may include a time for which the door **21** or **22** is kept open exceeding a first defined time (e.g., about 10 seconds).

[0320] In response to a lapse of the first defined time without detecting of closing of the door **21** or **22** after the opening of the door **21** or **22** is detected by the door sensor **343**, the controller **350** may determine that a defined condition associated with opening of the door **21** or **22** is satisfied.

[0321] In another example, the defined condition associated with opening of the door **21** or **22** may include a cumulative time of opening the door **21** or **22** exceeding a second defined time (e.g., about 1 minute).

[0322] The controller **350** may count a cumulative time from when opening of the door **21** or **22** is detected by the door sensor **343** to when closing of the door **21** or **22** is detected, and determine that the defined condition associated with opening of the door **21** or **22** is satisfied when the counted time (the cumulative time for which the door **21** or **22** is open) exceeds the second defined time (e.g., about 1 minute).

[0323] In another example, the defined condition associated with opening of the door **21** or **22** may include heat capacity of an object stored in the storage **11** after the door **21** or **22** is opened being

larger than a defined size.
[0324] The controller **350** may turn on a camera (e.g., an infrared camera) for capturing an image of the inside of the storage **11** when opening of the door **21** or **22** is detected by the door sensor **343**, identify heat capacity of an object placed in the storage **11** after the door **21** or **22** is opened based on the image obtained from the camera that captures the image of the inside of the storage **11**, and determine that the defined condition associated with the opening of the door **21** or **22** is satisfied when the identified heat capacity of the object is larger than the defined size.
[0325] Based on the defined condition associated with opening of the door **21**, **22**, **23** or **24** being satisfied in **2100**, the controller **350** may start the cooling mode in **2200**.
[0326] In the disclosure, the cooling mode corresponds to a mode for obtaining a predicted temperature value of the storage **11**, and determining whether to operate the thermoelectric element **530** based on the predicted temperature value of the storage **11**. In the disclosure, when it is determined that operation of the thermoelectric element **530** is not required even when the cooling mode is started, the cooling mode may be terminated without operation of the thermoelectric element **530**, and when it is determined that operation of the thermoelectric element is required, the cooling mode may be terminated as the operation of the thermoelectric element **530** is stopped after operation of the thermoelectric element **530**.
[0327] Based on the start of the cooling mode in **2200**, the controller **350** may initialize values associated with the defined condition associated with opening of the door.
[0328] For example, the defined condition associated with a door opening time may include a cumulative time of opening the door exceeding a defined time, and the controller **350** may initialize the cumulative time based on the start of the cooling mode.
[0329] Based on the start of the cooling mode in **2200**, the controller **350** may obtain a predicted temperature value of the storage **11** in **2300** by inputting sensor data collected by the sensor module **340** to a temperature prediction model.
[0330] As described above, the controller **350** may obtain the predicted temperature value of the storage **11** by inputting the sensor data to the temperature prediction model in various methods.
[0331] In an embodiment, to reduce the data processing load, the controller **350** may obtain the predicted temperature value of the storage **11** by inputting the sensor data to the temperature prediction model at defined intervals (e.g., of 5 minutes).
[0332] However, when a rapid temperature change in the storage **11** is expected, there is a need to quickly check the predicted temperature value of the storage **11** by changing the defined interval.
[0333] In an embodiment, the controller **350** may change the defined interval.
[0334] For example, the controller **350** may change the defined interval based on the temperature of the storage **11**. The controller **350** may change the defined interval based on a temperature change value of the storage **11** for each unit time. The controller **350** may linearly or non-linearly adjust the defined interval to be shorter as the temperature change value of the storage **11** for each unit time increases.
[0335] According to the disclosure, provided is the refrigerator **1** capable of obtaining more accurate predicted temperature values by obtaining the predicted temperature value of the storage **11** at relatively short intervals when there is a large temperature change in the storage **11**.
[0336] In another example, the controller **350** may change the defined interval based on the difference between the predicted temperature value and the target temperature value. The controller **350** may linearly or non-linearly adjust the defined interval to be shorter as the difference between the predicted temperature value and the target temperature value increases.
[0337] According to the disclosure, provided is the refrigerator **1** capable of obtaining more accurate predicted temperature values by obtaining the predicted temperature value of the storage **11** at relatively short intervals when the temperature change in the storage **11** is expected to be large.
[0338] The controller **350** may compare the predicted temperature value of the storage **11** with the

target temperature of the storage **11**, in **2400**.

[0339] Specifically, the controller **350** may determine whether the predicted temperature value of the storage **11** is larger than the target temperature value of the storage **11** by a defined value. In other words, the controller **350** may determine whether the difference between the predicted temperature value of the storage **11** and the target temperature value of the storage **11** is larger than the defined value (e.g., 10° C.).

[0340] For convenience of explanation, the difference between the predicted temperature value of the storage **11** and the target temperature value of the storage **11** will now be referred to as a difference value.

[0341] As described above, the controller **350** may determine the target temperature based on a set temperature for the storage **11** and sensor data outside the refrigerator collected by the outdoor sensor **342**. For example, the controller **350** may determine the target temperature to be lower than the set temperature for the storage **11** when the temperature and/or humidity outside the refrigerator is high.

[0342] FIG. **11** illustrates an example in which the thermoelectric element **530** is not operated even after a refrigerator starts a cooling mode, according to an embodiment.

[0343] Referring to FIG. **11**, the controller **350** may terminate the cooling mode without operating the thermoelectric element **530** based on the difference value being maintained to be less than a defined value T**1** (no in **2400** or yes in **2450**) until the refrigeration cycle is performed as many as the defined number of times (e.g., two times) from time to at which the cooling mode is started.

[0344] In other words, the controller **350** may terminate the cooling mode in **2700** without operation of the thermoelectric element **530** based on the difference value being equal to or less than the defined value T**1** until the refrigeration cycle is performed as many as the defined number of times.

[0345] The terminating of the cooling mode by the controller **350** is because the difference value maintained to be less than the defined value even though the refrigeration cycle has been performed as many as the defined number of times means there is no longer a room for the temperature of the storage **11** to rise according to an event associated with opening the door **21** or **22**.

[0346] While the refrigeration cycle is being performed, the controller **350** may count the number of times that the refrigeration cycle is performed inclusive of the refrigeration cycle that is being currently performed.

[0347] When the refrigeration cycle is not performed, the controller **350** may count the number of refrigeration cycles that will be performed subsequently.

[0348] For example, the controller **350** may count the number of times that the refrigeration cycle is performed based on the compressor **2** being changed from an operation state to a terminated state.

[0349] The controller **350** may terminate the cooling mode at time t**2** at which the difference value is maintained to be equal to or less than the defined value and the number of times that the refrigeration cycle is performed exceeds the defined number of times.

[0350] In various embodiments, the controller **350** may terminate the cooling mode without operation of the thermoelectric element **530** based on a lapse of a reference time while the difference value is maintained to be equal to and less than the defined value.

[0351] In various embodiments, the controller **350** may terminate the cooling mode without operation of the thermoelectric element **530** based on a gradient of the difference value being changed from a positive value to a negative value while the difference value is maintained to be equal to and less than the defined value.

[0352] According to the disclosure, when the thermoelectric element **530** does not have to be operated to maintain the temperature of the storage **11**, the cooling mode may be terminated without operation of the thermoelectric element **530**, thereby reducing energy consumption.

[0353] As shown in FIG. **11**, when the cooling mode begins while the refrigeration cycle is being performed by the compressor **2**, the predicted temperature of the storage **11** may not rise significantly. However, when opening time of the door **21** or **22** is significantly long or an object with significantly high heat capacity is put in the storage **11** even after the cooling mode begins while the refrigeration cycle is being performed by the compressor **2**, there is a room for the predicted temperature of the storage **11** to rise significantly.

[0354] Based on the difference value being larger than the defined value in **2400** while operating in the cooling mode, the controller **350** may operate the thermoelectric element **530** in **2500**.

[0355] A condition in which the difference value is larger than the defined value while operating in the cooling mode may correspond to the condition M**1** as described in FIG. **9**.

[0356] When the difference value is larger than the defined value while operating in the cooling mode, the controller **350** may operate the thermoelectric element **530** with the first control parameter M**1**. The first control parameter may include a lookup table in which difference values match on/off duty ratios of the thermoelectric element **530**.

[0357] When the difference value is larger than the defined value, the controller **350** may control the duty ratio of the thermoelectric device **530** based on the difference value. For example, when the difference value is larger than the defined value, the controller **350** may control the duty ratio of the thermoelectric device **530** to be higher the larger the difference value.

[0358] In an embodiment, when a control condition with higher priority than the cooling mode is satisfied and thus the thermoelectric element **530** is controlled with a second control parameter even though the difference between the predicted temperature value and the target temperature value is larger than the defined value while operating in the cooling mode, the controller **350** may keep controlling the thermoelectric element **530** based on the second control parameter.

[0359] The control condition with higher priority than the cooling mode may include the control condition L**1** having higher priority than the control condition M**1** corresponding to the cooling mode.

[0360] Specifically, when the thermoelectric element **530** is controlled according to the control parameter L**2** as the condition L**1** is satisfied even when the condition M**1** is satisfied, the controller **350** may keep controlling the thermoelectric element **530** based on the control parameter L**2**.

[0361] For example, the controller **350** may not operate the thermoelectric element **530** when the thermoelectric element **530** is in an off state according to the condition L**1** even when the difference value is larger than the defined value while operating in the cooling mode.

[0362] In another example, the controller **350** may operate the thermoelectric element **530** with a first duty ratio when the difference value is larger than the defined value while operating in the cooling mode, but when the thermoelectric element **530** is operated with a second duty ratio according to the condition L**1** even when the difference value is larger than the defined value while operating in the cooling mode, the controller **350** may keep operating the thermoelectric element **530** with the second duty ratio.

[0363] In an embodiment, the controller **350** may operate the thermoelectric element **530** based on the difference value being larger than the defined value while operating in the cooling mode regardless of whether the compressor **2** operates (whether the refrigeration cycle proceeds).

[0364] Accordingly, the controller **350** may operate the thermoelectric element **530** while the refrigeration cycle is being progressed, or operate the thermoelectric element **530** when the refrigeration cycle is not progressed, or start the refrigeration cycle while operating the thermoelectric element **530**.

[0365] In an embodiment, the controller **350** may terminate the cooling mode in **2700** by stopping operation of the thermoelectric element **530** based on the difference value falling to or below a reference value in **2600** after operating the thermoelectric element **530**. The reference value may be smaller than the defined value and larger than the target temperature value.

[0366] FIG. **12** illustrates an example in which a compressor and a thermoelectric element are operated together when a refrigerator starts a cooling mode, according to an embodiment.

[0367] Referring to FIG. **12**, the controller **350** may start the cooling mode at time t**0** at which a defined condition associated with door opening time is satisfied while the refrigeration cycle is not progressed.
[0368] Afterward, as the temperature of the storage **11**, **12** or **13** rises, the refrigeration cycle may begin, and the difference value may exceed the defined value T**1** at time t**1** after or before the refrigeration cycle begins.
[0369] Based on the difference value exceeding the defined value T**1**, the thermoelectric element **530** may be operated; when the thermoelectric element **530** is operated while the refrigeration cycle is being performed, both the compressor **2** and the thermoelectric element **530** may be operated; when the thermoelectric element **530** is operated while the refrigeration cycle is not performed, only the thermoelectric element **530** may be operated.
[0370] In other words, the controller **350** may start the refrigeration cycle based on the cooling condition being satisfied, and based on the difference between the predicted temperature value and the target temperature value exceeding the defined value while performing the refrigeration cycle, the controller **350** may operate the compressor **2** and the thermoelectric element **530** together by operating the thermoelectric element **530**.
[0371] Afterward, the controller **350** may terminate the cooling mode by stopping operation of the thermoelectric element **530** at time t**2** at which the difference value falls to or below the reference value T**2**.
[0372] FIG. **13** illustrates an example in which only a thermoelectric element is operated among a compressor and the thermoelectric element when a refrigerator starts a cooling mode, according to an embodiment.
[0373] Referring to FIG. **13**, the controller **350** may start the cooling mode at time t**0** at which a defined condition associated with door opening time is satisfied while the refrigeration cycle is being progressed.
[0374] Afterward, even while the refrigeration cycle is being progressed according to an event associated with the door opening time, the difference value may exceed the defined value T**1** at time t**1**.
[0375] Even when the temperature of the storage **11**, **12** or **13** drops after the completion of the refrigeration cycle, the thermoelectric element **530** may be operated based on the difference value exceeding the defined value T**1**.
[0376] In other words, even though the temperature of the storage **11**, **12** or **13** is maintained at or below the target temperature, when the predicted temperature value of the storage **11** is high, only the thermoelectric element **530** may be operated among the compressor **2** and the thermoelectric element **530**.
[0377] For example, based on the difference value exceeding the defined value while the refrigeration cycle is not performed, the controller **350** may operate only the thermoelectric element **530** among the compressor **2** and the thermoelectric element **530** by operating the thermoelectric element **530**.
[0378] Afterward, the controller **350** may terminate the cooling mode by stopping operation of the thermoelectric element **530** at time t**2** at which the difference value falls to or below the reference value T**2**.
[0379] When the temperature of the storage **11**, **12** or **13** satisfies a cooling condition before the time t**2** is reached, it is obvious that the controller **350** may operate the thermoelectric element **530** and the compressor **2** together by operating the compressor **2**.
[0380] In an embodiment of the disclosure, by managing the future temperature of the storage **11** through operation of the thermoelectric element **530** and managing the current temperature of the storage **11**, **12** or **13** through operation of the compressor **2**, the temperature of the storage **11**, **12** or **13** may be maintained with minimum energy consumption.
[0381] In the meantime, when the difference value significantly increases during operation in the

cooling mode, the temperature of the storage **11** may not be maintained only by operating the thermoelectric element **530**.

[0382] In various embodiments, the controller **350** may operate the compressor **2** when the difference value exceeds a maximum set value while operating in the cooling mode.

[0383] That is, the controller **350** may determine whether to operate the compressor **2** based on the difference value.

[0384] FIG. **14** illustrates an example in which a compressor is operated while a thermoelectric element is operating when a refrigerator starts a cooling mode, according to an embodiment.

[0385] Referring to FIG. **14**, the controller **350** may start the cooling mode at time t**0** at which a defined condition associated with door opening time is satisfied after completion of the refrigeration cycle.

[0386] When the defined condition associated with the door opening time is satisfied right after the completion of the refrigeration cycle, it is more likely that the predicted temperature value rises sharply. With the sharp rise of the predicted temperature value, the difference value may exceed the defined value T**1** at time t**1**.

[0387] The thermoelectric element **530** may be operated based on the difference value exceeding the defined value T**1**, but the difference value may rise sharply and exceed the maximum set value T**3**.

[0388] The compressor **2** may be operated at time ta at which the difference value reaches the maximum set value T**3**.

[0389] However, to prevent excessive energy consumption, the compressor **2** may be stopped at time tb at which the difference value falls to a defined value which is smaller than the maximum set value T**3** and larger than the defined value T**1**.

[0390] Afterward, the thermoelectric element **530** may be stopped at time t**2** at which the difference value drops to the reference value T**2**.

[0391] The controller **350** may operate the compressor **2** based on the difference value reaching the maximum set value T**3**. For example, the controller **350** may operate the compressor **2** based on the difference value reaching the maximum set value T**3** regardless of whether the cooling condition is satisfied.

[0392] The controller **350** may stop operating the compressor **2** based on the difference value falling to or below a defined value after operation of the compressor **2**. The defined value may be set to a value between the maximum set value T**3** and the defined value T**1** in advance.

[0393] The controller **350** may terminate the cooling mode by stopping operation of the thermoelectric element **530** based on the difference value falling to or below the reference value T**2** after the compressor **2** is stopped.

[0394] According to the disclosure, rapid temperature changes in the storage **11** may be proactively dealt with by cooling the storage **11** in advance by using not only the thermoelectric element **530** but also the compressor **2** when the predicted temperature of the storage **11** rises sharply.

[0395] According to an embodiment of the disclosure, the refrigerator **1** includes the main body **100** defining the storage **11**, **12** or **13**; the door **21**, **22**, **23** or **24** opening or closing the storage **11**, **12** or **13**; the refrigeration cycle device **450** including the compressor **2** and the evaporator **3** and configured to cool the storage **11**, **12** or **13**; the thermoelectric element **530** configured to cool the storage **11**, **12** or **13**; at least one sensor **340** configured to collect sensor data associated with the refrigerator **1**; and at least one processor **351** or **361** configured to perform a refrigeration cycle by operating the compressor **2** based on a cooling condition being satisfied, and start a cooling mode based on a defined condition associated with an opening time of the door **21**, **22**, **23** or **24** being satisfied, wherein the at least one processor **351** or **361** is configured to, based on the start of the cooling mode, obtain a predicted temperature value of the storage **11**, **12** or **13** by inputting the sensor data to a temperature prediction model, and operate the thermoelectric element **530** based on a difference between the predicted temperature value and a target temperature value being larger

than the defined value T**1**.
[0396] The at least one processor **351** or **361** may be configured to terminate the cooling mode by stopping operation of the thermoelectric element **530** in response to a difference between the predicted temperature value and the target temperature value falling to or below the reference value T**2** after the operating of the thermoelectric element **530**.
[0397] The at least one processor **351** or **361** may be configured to terminate the cooling mode without operation of the thermoelectric element **530** based on a difference between the predicted temperature value and the target temperature value being equal to or less than the defined value T**1** until the refrigeration cycle is performed as many as a defined number of times.
[0398] The at least one processor **351** or **361** may be configured to determine the target temperature value based on a set temperature and the sensor data.
[0399] The at least one processor **351** or **361** may be configured to operate the compressor **2** and the thermoelectric element **530** together by operating the thermoelectric element **530** based on a difference between the predicted temperature value and the target temperature value exceeding the defined value T**1** while performing the refrigeration cycle.
[0400] The at least one processor **351** or **361** may be configured to operate only the thermoelectric element **530** among the compressor **2** and the thermoelectric element **530** by operating the thermoelectric element **530** based on a difference between the predicted temperature value and the target temperature value exceeding the defined value T**1** while the refrigeration cycle is not performed.
[0401] The at least one processor **351** or **361** may be configured to operate the thermoelectric element **530** based on a first control parameter in response to a difference between the predicted temperature value and the target temperature value being larger than the defined value T**1**.
[0402] The at least one processor **351** or **361** may be configured to keep controlling the thermoelectric element **530** based on a second control parameter, in response to the thermoelectric element **530** being controlled with the second control parameter as a control condition having higher priority than the cooling mode is satisfied even though the difference between the predicted temperature value and the target temperature value is larger than the defined value T**1** while operating in the cooling mode.
[0403] The at least one processor **351** or **361** may be configured to obtain the predicted temperature value of the storage **11**, **12** or **13** at defined intervals.
[0404] The at least one processor **351** or **361** may be configured to change the defined interval based on temperature of the storage **11**, **12** or **13**.
[0405] The at least one processor **351** or **361** may be configured to change the defined interval based on a difference between the predicted temperature value and the target temperature value.
[0406] The at least one processor **351** or **361** may include a first processor **351** configured to control the compressor **2** and the thermoelectric element **530**; and a second processor **361** configured to obtain a predicted temperature value of the storage **11**, **12** or **13**.
[0407] The first processor **351** may be configured to instruct the second processor **361** to perform the temperature prediction model in response to the start of the cooling mode, and the second processor **361** may be configured to obtain the predicted temperature value in response to receiving the instruction from the first processor **351** and send the predicted temperature value to the first processor **351**.
[0408] The at least one processor **351** or **361** may be configured to control a duty ratio of the thermoelectric element **530** based on a difference between the predicted temperature value and the target temperature value.
[0409] The at least one processor **351** or **361** may be configured to determine whether to operate the compressor **2** based on a difference between the predicted temperature value and the target temperature value.
[0410] The defined condition may include a cumulative time of opening the door **21**, **22**, **23** or **24**

exceeding a defined time.

[0411] The at least one processor **351** or **361** may initialize the cumulative time based on the start of the cooling mode.

[0412] According to an embodiment of the disclosure, a method for controlling the refrigerator **1** may include performing a refrigeration cycle by operating the compressor **2** based on a cooling condition being satisfied; starting a cooling mode based on a defined condition associated with an opening time of the door **21**, **22**, **23** or **24** for opening or closing the storage **11**, **12** or **13** being satisfied; based on the start of the cooling mode, obtaining a predicted temperature value of the storage **11**, **12** or **13** by inputting sensor data associated with the refrigerator **1** to a temperature prediction model, and operating the thermoelectric element **530** for cooling the storage **11**, **12** or **13** based on a difference between the predicted temperature value and a target temperature value being larger than the defined value T**1**.

[0413] The method may further include terminating the cooling mode by stopping operation of the thermoelectric element **530** in response to a difference between the predicted temperature value and the target temperature value falling to or below the reference value T**2** after the operating of the thermoelectric element **530**.

[0414] The method may further include terminating the cooling mode based on a difference between the predicted temperature value and the target temperature value being equal to or less than the defined value T**1** until the refrigeration cycle is performed as many as a defined number of times.

[0415] The operating of the thermoelectric element **530** may include operating the compressor **2** and the thermoelectric element **530** together by operating the thermoelectric element **530** based on a difference between the predicted temperature value and the target temperature value exceeding the defined value T**1** while performing the refrigeration cycle.

[0416] The operating of the thermoelectric element **530** may include operating only the thermoelectric element **530** among the compressor **2** and the thermoelectric element **530** by operating the thermoelectric element **530** based on a difference between the predicted temperature value and the target temperature value exceeding the defined value T**1** while the refrigeration cycle is not performed.

[0417] The operating of the thermoelectric element **530** may include operating the thermoelectric element **530** based on a first control parameter, and the method may further include keeping controlling the thermoelectric element **530** based on a second control parameter, in response to the thermoelectric element **530** being controlled with the second control parameter as a control condition having higher priority than the cooling mode is satisfied even though the difference between the predicted temperature value and the target temperature value is larger than the defined value T**1** while operating in the cooling mode.

[0418] The defined condition may include a cumulative time of opening the door **21**, **22**, **23** or **24** exceeding a defined time, and the method may further include initializing the cumulative time based on the start of the cooling mode.

[0419] Meanwhile, the embodiments of the disclosure may be implemented in the form of a recording medium for storing instructions to be carried out by a computer. The instructions may be stored in the form of program codes, and when executed by a processor, may generate program modules to perform operations in the embodiments of the disclosure. The recording media may correspond to computer-readable recording media.

[0420] The computer-readable recording medium includes any type of recording medium having data stored thereon that may be thereafter read by a computer. For example, it may be a read only memory (ROM), a random access memory (RAM), a magnetic tape, a magnetic disk, a flash memory, an optical data storage device, etc.

[0421] The computer-readable storage medium may be provided in the form of a non-transitory storage medium. The term ‘non-transitory storage medium’ may mean a tangible device without

including a signal, e.g., electromagnetic waves, and may not distinguish between storing data in the storage medium semi-permanently and temporarily. For example, the non-transitory storage medium may include a buffer that temporarily stores data.
[0422] In an embodiment of the disclosure, the aforementioned method according to the various embodiments of the disclosure may be provided in a computer program product. The computer program product may be a commercial product that may be traded between a seller and a buyer. The computer program product may be distributed in the form of a recording medium (e.g., a compact disc read only memory (CD-ROM)), through an application store (e.g., Play store™), directly between two user devices (e.g., smart phones), or online (e.g., downloaded or uploaded). In the case of online distribution, at least part of the computer program product (e.g., a downloadable app) may be at least temporarily stored or arbitrarily created in a recording medium that may be readable to a device such as a server of the manufacturer, a server of the application store, or a relay server.
[0423] The embodiments of the disclosure have thus far been described with reference to accompanying drawings. It will be obvious to those of ordinary skill in the art that the disclosure may be practiced in other forms than the embodiments of the disclosure as described above without changing the technical idea or essential features of the disclosure. The above embodiments of the disclosure are only by way of example, and should not be construed in a limited sense.

Claims

1. A refrigerator comprising: a main body defining a storage chamber; a door configured to open and close the storage chamber; a refrigeration cycle device including a compressor and an evaporator and operable to cool the storage chamber; a thermoelectric element operable to cool the storage chamber; at least one sensor configured to produce sensor data associated with the refrigerator; and at least one processor configured to: operate the compressor to perform a refrigeration cycle based on a cooling condition being satisfied, start a cooling mode based on a defined condition associated with a time that the door is open being satisfied, and based on the cooling mode being started, obtain a predicted temperature value of the storage chamber from a temperature prediction model based on the sensor data produced by the at least one sensor, and operate the thermoelectric element to cool the storage chamber based on a difference between the predicted temperature value obtained from the temperature prediction model and a target temperature value being greater than a defined value.
2. The refrigerator of claim 1, wherein the at least one processor is further configured to terminate the cooling mode by stopping operating the thermoelectric element in response to the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to a reference value after operating the thermoelectric element to cool the storage chamber.
3. The refrigerator of claim 1, wherein the at least one processor is further configured to terminate the cooling mode without operating the thermoelectric element based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to the defined value until the refrigeration cycle is performed a defined number of times.
4. The refrigerator of claim 1, wherein the at least one processor is further configured to determine the target temperature value based on a set temperature and the sensor data produced by the at least one sensor.
5. The refrigerator of claim 1, wherein the at least one processor is further configured to operate the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is being performed such that the

thermoelectric element and the compressor are being operated together.
6. The refrigerator of claim 1, wherein the at least one processor is further configured to operate the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is not being performed such that only the thermoelectric element is being operated from among the thermoelectric element and the compressor.
7. The refrigerator of claim 1, wherein the at least one processor is further configured to: operate the thermoelectric element to cool the storage chamber based on a first control parameter in response to the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value, and while operating the thermoelectric element based on a second control parameter in response to a control condition having higher priority than the cooling mode being satisfied, even though the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value is greater than the defined value, keep operating the thermoelectric element to cool the storage chamber based on the second control parameter.
8. The refrigerator of claim 1, wherein the at least one processor is further configured to: obtain the predicted temperature value of the storage chamber from the temperature prediction model at defined intervals, and change the defined intervals based on a temperature of the storage chamber.
9. The refrigerator of claim 1, wherein the at least one processor is further configured to: obtain the predicted temperature value of the storage chamber from the temperature prediction model at defined intervals, and change the defined intervals based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.
10. The refrigerator of claim 1, wherein the at least one processor includes: a first processor configured to control the refrigeration cycle device and the thermoelectric element; and a second processor configured to obtain the predicted temperature value of the storage chamber from the temperature prediction model; and the first processor is further configured to send to the second processor an instruction to obtain the predicted temperature value from the temperature prediction model in response to the cooling mode being started, and the second processor is further configured to: obtain the predicted temperature value from the temperature prediction model in response to the instruction of the first processor being received, and send the predicted temperature value obtained from the temperature prediction model to the first processor.
11. The refrigerator of claim 1, wherein the at least one processor is further configured to control a duty ratio of the thermoelectric element based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.
12. The refrigerator of claim 1, wherein the at least one processor is further configured to determine whether to operate the compressor based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.
13. The refrigerator of claim 1, wherein the defined condition includes a cumulative time that the door is open exceeding a defined time, and the at least one processor is further configured to initialize the cumulative time based on the cooling mode being started.
14. A method for controlling a refrigerator including a main body defining a storage chamber, a door configured to open and close the storage chamber, a refrigeration cycle device including a compressor and an evaporator and operable to cool the storage chamber, a thermoelectric element operable to cool the storage chamber, at least one sensor configured to produce sensor data associated with the refrigerator, and at least one processor, the method comprising: by the at least one processor, operating the compressor to perform a refrigeration cycle based on a cooling condition being satisfied, starting a cooling mode based on a defined condition associated with a time that the door is open being satisfied, and based on the cooling mode being started, obtaining a

predicted temperature value of the storage chamber from a temperature prediction model based on the sensor data produced by the at least one sensor, and operating a thermoelectric element to cool the storage chamber based on a difference between the predicted temperature value obtained from the temperature prediction model and a target temperature value being greater than a defined value.

15. The method of claim 14, further comprising: by the at least one processor, terminating the cooling mode by stopping operating the thermoelectric element in response to the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to a reference value after operating the thermoelectric element to cool the storage chamber.

16. The method of claim 14, further comprising: by the at least one processor, terminating the cooling mode without operating the thermoelectric element based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being less than or equal to the defined value until the refrigeration cycle is performed a defined number of times.

17. The method of claim 14, further comprising: by the at least one processor, determining the target temperature value based on a set temperature and the sensor data produced by the at least one sensor.

18. The method of claim 14, wherein the operating the thermoelectric element comprises: operating the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is being performed such that the thermoelectric element and the compressor are being operated together.

19. The method of claim 14, wherein the operating the thermoelectric element comprises: operating the thermoelectric element to cool the storage chamber based on the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value being greater than the defined value while the refrigeration cycle is not being performed such that only the thermoelectric element is being operated from among the thermoelectric element and the compressor.

20. The method of claim 14, further comprising: by the at least one processor, obtaining the predicted temperature value of the storage chamber from the temperature prediction model at defined intervals; and changing the defined intervals based on at least one of a temperature of the storage chamber or the difference between the predicted temperature value obtained from the temperature prediction model and the target temperature value.